



A review and critical analysis of multimodal datasets for emotional AI

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Abstract

With the increasing interest in digital technologies, emotion recognition plays an important role in several applications such as healthcare computer-aided diagnosis, social media analysis, opinion mining and recommendation systems, understanding human behavior and interaction in workplaces, effective communication and linguistic analysis, and cognitive human–machine interaction. This field is receiving a growing interest in recent years. In this paper, we present a thorough review of emotional artificial intelligence through identification and in-depth analysis of existing multimodal datasets along with their related research directions and methodologies. It establishes essential requirements for the development of a multimodal dataset and outlines challenges spanning its entire lifecycle, from recording to deployment. Moreover, a taxonomy of various categories and applications is introduced based on the key characteristics of various multimodal datasets. Finally, the paper concludes with discussions and insights into future directions and prospects for standard schemes to facilitate the efficient development of reliable and reusable benchmark datasets that can help researchers and developers advance this field.

Keywords Affective computing · Opinion mining · Emotional AI · Multimodal learning · Datasets · Video analytics

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1 Introduction

Emotional AI or affective computing refers to the field of artificial intelligence that focuses on recognizing, understanding, interpreting, responding, or simulating human feelings, intentions, or reactions to various situations (Zhao et al. 2022a). Affect state is provoked by various internal and external stimuli and can have a crucial impact on person's life and interactions with others. Affective computing technology is significantly important for understanding human behavior and enabling social sensing. It is also helpful in healthcare such as tracking mood changes of patients undergoing mental treatment. Moreover, it enables building systems and machines, such as computers, robots, or smartphones, to detect, understand, and react to human emotional states in a natural, personalized, and empathetic manner (Chakriswaran et al. 2019).

The research community's profound interest in emotional AI domains is fueled by various factors, owing to their extensive applications across diverse fields like social media analysis (Chandrasekaran et al. 2021), online marketing and brand analysis (Hu and Liu 2004; Pusztahelyi et al. 2020), expressive text-to-speech synthesis (Alm et al. 2005), conversational AI systems, voice-based assistants and chatbots (Zhang et al. 2024a; Park et al. 2023; Teye et al. 2022; Poria et al. 2019; Urquhart et al. 2022), AI-enabled toys and social robots (Song et al. 2023; Dhiman et al. 2023), healthcare (Cano et al. 2023; Hasnul et al. 2021b; Denecke and Deng 2015; Zhou et al. 2023), smart transportation and safe driving (Zepf et al. 2020), educational environments and learner's engagement (Manolev et al. 2020; Keerthika et al. 2024), and surveillance and security (Butt et al. 2023; Kaushik et al. 2022).

For example, on social network platforms, users often contribute content in different formats including text, audio, images, and videos. The user-generated content might include opinions, attitudes, reviews, and sentiments toward entities including products, organizations, individuals, and services. This encourages the development of accurate and useful tools for analyzing the rapid growth of user-generated content. Various tools have been built to track public viewpoints and report acceptable performance. However, text-based tools are still inadequate to capture sentiments and detect emotions that are not overtly conveyed (Cambria 2016), e.g. due to language dependency, domain dependency, language ambiguity (Mantello et al. 2023), sarcasm (Kumar et al. 2023), informal style of writing, and affective ambiguity (Alvarez-Gonzalez et al. 2021). Similarly, audio-based emotional AI is suffering from issues related to insufficient representation, low voice quality, background noise, and the disposition of voice recording devices. This is also true for visual-based emotional AI which has limitations related to illumination variations, occlusions, non-frontal head poses, identity or gender biases,

low-intensity expressions, and variability of facial expressions. On the other hand, the main drawbacks of physiological-based emotional AI include its complexity as well as the cost of data collection and affect detection.

By fusing data from various modalities, such as text, voice tone, facial expressions, gestures, and physiological signals, multimodal emotional AI can achieve higher accuracy (Wang et al. 2024) and robust performance (Zhu et al. 2024), especially in complex scenarios where an emotional expression varies across different modalities. This approach can handle situations where one modality may not provide a complete picture of the speaker's emotions or may be ambiguous. There are a variety of sensors that can detect emotional

signals and cues such as facial expressions and tears using video cameras, body postures using wearable sensors, eye movement using an eye-tracking sensors, voice intonation using microphones, skin temperature and sweat using Galvanic Skin Response (GSR) or Electrodermal Activity (EDA) sensors, heart rate using Electrocardiography (ECG) sensors, brain activity using Electroencephalography (EEG) sensors, blood oxygen monitors, or respiratory rate using chest bands or capnography noninvasive devices. As a result, emotional AI can be categorized into distinct types based on the data source it leverages (Cai et al. 2023). These categories encompass emotional text-focused AI, which involves the analysis of word choices and linguistic patterns using Natural Language Processing (NLP); audio-focused AI, which centers on the examination of voice intonation; visual-focused AI, which concentrates on the analysis of facial expressions and body postures; and physiological-focused AI, which involves the scrutiny of physiological signals, such as EEG, ECG, or EDA (Lin and Li 2023; Garg et al.).

1.1 Motivation and scope of contributions

Despite the recent development, emotional AI still faces many unresolved challenges. This necessitates the need for exploring new innovative methods to develop systems that are more dependable and efficient. The success of these endeavors is greatly affected by the availability of datasets. Although there are several unimodal datasets, emotion recognition is generally a challenging task, and recent trends attempt to combine multimodal signals to provide more accurate representations and recognition results (Rahate et al. 2022; Ahmed et al. 2023; Fu et al. 2023). To further motivate the current trend for multimodal datasets and multimodal machine learning that incorporate auditory, visual, textual, and physiological modalities, we begin with an analysis of the strengths and limitations of each individual modality considered in this study. We also present a sample of corresponding resources, including related reviews and common datasets, as shown in Table 1. It is important to note that there are multiple prominent theories attempting to understand how and why people experience emotion and related behavior from psychological and cognitive perspective. However, these theories are not discussed in this paper and we refer interested reader to other such as (Izard 2009; Cowen and Keltner 2021; Keltner et al. 2023; Schiller et al. 2024). The main contributions of this paper are to:

- Explore and identify major trends in contemporary multimodal emotional AI applications and survey available datasets, providing valuable insights into the evolving landscape of this field.
- Thoroughly describe the key characteristics of multimodal datasets and offer a comprehensive taxonomy of the diverse elements involved in these datasets.
- Explain the requirements and the process of building multimodal datasets and identify the main challenges and obstacles associated with this process,
- Outline challenges in reproducing and deploying multimodal datasets, offering a road map for researchers while exploring ethical concerns in emotional AI research.
- Provide a critical analysis of solutions and approaches employed in the field, contributing to the advancement of best practices.

Table 1 Resources, related reviews, strengths, and limitations of individual modalities (where SA: sentiment analysis, ER: Emotion recognition, PA: pain assessment, PT: personality traits, MT: medical treatment)

Affect Modality	Datasets	Applications	Related Reviews	Strengths	Limitations
Text (Written language)	Sentiment140 (Go et al. 2009), SemEval (Pontiki et al. 2015, 2016; Acheampong et al. 2020), ASC-QA (Wang et al. 2019), MAMS (Jiang et al. 2019), ARTS (Xing et al. 2020), Amazon Review (Ni et al. 2019), KurdiSent (Badawi et al. 2024)	SA: Polarity detection, review analysis, business analysis, stock price prediction, Social media monitoring, Voice of customers	(Wankhade et al. 2022; Chan et al. 2023; Zhou and Ye 2023; Bensoltane and Zaki 2023; Phan et al. 2023)	Precise, permanence, wide reach, cost-effectiveness, explicit expression, content searchability	Language dependency, domain dependency, language ambiguity, sarcasm, informal style of writing, Grammatical errors, affective ambiguity
Audio (spoken language, tone of voice, laughing, crying, etc.)	MPEG-4 (Schuller et al. 2005), TESS (Pichora-Fuller and Dupuis 2020), EmoDB (Burkhardt et al. 2005), CASIA (Zhang and Jia 2008), BHUDES (Mao et al. 2009), EESDB (Wang et al. 2014a)	ER, SA	(Akçay and Oğuz 2020; Atmaja et al. 2022; Kumar et al. 2022)	More natural and spontaneous, less-private	Insufficient representation, poor voice quality, ambient noise, and the placement of voice recording devices
Visual (facial expressions, body gestures/postures, eye movement, etc)	CK+ (Lucey et al. 2010), KVDERW (Khanh et al. 2021) KVDERW-Eng (Khanh et al. 2021) MMI (Pantic et al. 2005) AFEW (Dhall et al. 2012), FER-2013 (Goodfellow et al. 2013)	ER, MT	(Li and Deng 2020a; Costa et al. 2023)	Rich and nuanced, technological integration, useful for limited verbal communication, natural social interaction	Variations in lighting, occlusions, non-frontal head poses, bias related to identity, the recognition of low-intensity expressions, facial expression variability
Physiological (EEG, ECG, etc.)	GAMEEMO (Alakus et al. 2020), DEAP (Koelstra et al. 2012), (Hasnul et al. 2021a; Bagirathan et al. 2021; Hsu et al. 2017)	ER, MT, PA, PT	(Lin and Li 2023; Garg et al.; Ba and Hu 2023; Garg et al.; Hullyyah et al. 2017)	More robust for social masking avoidance, less sensitive to false emotions and fake feelings	Complexity for data collection, cost

1.2 Organization of the paper

The structure of this paper is as follows: Sect. 1 provides an overview of the emotional AI, defines the key contributions, and outlines the structure of the manuscript. Section 2 outlines the objectives of the review, specifies the research questions, and presents the details of the review methodology. Section 3 surveys existing multimodal datasets and categorizes them based on their applications and tasks. Section 4 provides a taxonomy and in-depth discussion of the characteristics and major trends of multimodal datasets for emotional AI. Section 5 describes the workflow, key requirement and challenges for developing multimodal emotional datasets. Section 6 discusses major emerging research directions in the field. Sec-

tion 7 summarizes the paper and highlight recommendations for developers of emotional multimodal datasets.

2 Review objectives and methodology

We followed a tailored review process designed to gain deeper insights about multimodal datasets by analyzing the information published about available datasets and information in other papers utilizing these datasets. We attempted to answer the following primary research questions:

- RQ1 Which multimodal datasets for emotional AI exist and what are their main characteristics and major trends?
- RQ2 What is the process of building a multimodal dataset?
- RQ3 What are the main requirements and challenges for developing a multimodal dataset and how they have been addressed by existing multimodal datasets?
- RQ4 What are the major issues obstructing the reproduction and deployment of such datasets?
- RQ5 What are the emerging research directions in the field?

As a starting step in our study, relevant publications have been collected from Google Scholar and the search process has been carried out in several iterations. We utilized search terms and queries related to (multimodal OR multi-modal OR bimodal), (sentiment OR emotion OR affect), (analysis OR detection OR recognition), and (dataset OR database). This resulted in a large number of papers. We then selected papers that are related to multimodal emotional AI. We also augmented this list with other papers found by searching on specific physiological traits and papers on multimodal deep learning for emotion recognition. We also checked Mendeley and Google Databases for any additional related work. Datasets that were eligible for *inclusion* in our study were restricted to:

- a) The dataset should be published in peer-reviewed journals or conferences, or archived papers between 2000 and 2023. This time frame was chosen because research prior to 2000 primarily focused on unimodal datasets while multimodal datasets have gained prominence in more recent years. Additionally, after submission of the first version of this paper, a number of datasets related to large vision-language models (LVLMs) were introduced in 2024 and 2025 and are discussed separately under emerging directions.
- b) dataset should incorporate at least two modalities (e.g. audio, visual, textual, physiological such as EEG, ECG, GSR) of the participating subjects.
- c) dataset should focus on emotional states such as those defined by Ekman's or Plutchik's emotion categories or sentiment labels or scores.

Therefore, datasets that did not meet any of the above criteria were excluded from our study. In addition, datasets primarily focused on psychological research related to emotional stimuli and cognitive reactions, such as (Diconne et al. 2021), were also excluded as they fall outside the scope of our machine learning-centric perspective on multimodal emotional AI datasets. Subsequently, related work in the literature can belong the follow-

ing three groups: (a) *MM-Type I*– papers that created multimodal datasets for emotional AI, (b) *MM-Type II*– collection of papers that used and evaluated multimodal datasets, and (c) *MM-Type III*– neither created datasets nor evaluated other datasets, but they described multimodal datasets for video opinion mining. We tracked the second and third types of papers (MM-Type II and MM-Type III) by looking at their references to find the main target papers in the first group, MM-Type I.

The following sections attempt to provide answers for the research questions of this study. Section 3 and 4 present answers for RQ1. Section 5 answers RQ2, RQ3 and RQ4. Section 6 answers RQ5.

3 Survey of multimodal datasets for emotional AI

Following the methodology outlined in the previous section, we identified a final list of 63 multimodal datasets that met the inclusion criteria. Of these, 49 datasets were homogeneous (i.e., they involved the same modalities across all data instances), while 14 datasets were heterogeneous (i.e., they involved a mix of different modalities for different data instances). On such datasets, more in-depth critical analysis and characterization have been performed in our study, which differs from other existing reviews in several aspects as outlined in Table 2. In the following subsections, we review several examples of the most popular ones for various tasks including sentiment polarity detection, emotion analysis and recognition, sarcasm detection, and demographic segmentation.

3.1 Multimodal polarity detection datasets

YouTube dataset (Morency et al. 2011): To our knowledge, the first exploration of sentiment analysis across three modalities was conducted by Morency and Mihalcea using a collection of 47 YouTube videos. Their findings suggested that this task is feasible and can benefit from the collaborative utilization of auditory, visual, and textual modalities. The videos in this dataset covered different topics expressed in English and were collected using the search keywords: opinion, review, best perfume, toothpaste, business, war, job, hate, and like. They were expressed by 20 male and 27 female speakers belonging to different cultures (e.g., Caucasian, African-American, Hispanic, Asian). Their ages ranged from 14 to 60 years, approximately. The videos were annotated manually as 13 positive, 12 negative, and 22 neutral videos, by three annotators with an average annotator agreement of 78.7%.

ICT-MMMO (Multi-Modal Movie Opinion) Dataset (Wöllmer et al. 2013): This is a multimodal dataset, composed of 370 movie review videos collected from YouTube and ExpoTV. Two search rounds were conducted to collect the dataset. The first collection was conducted on the YouTube website by search queries for movie review videos and opinions, including optionally the name of recent movies as listed by imdb.com. Out of the total 370 videos, 308 were retrieved and annotated as 228 positive, 23 neutral, and 57 negative. The second search round was conducted using the website ExpoTV.com to retrieve 62 negative movie review videos. Additionally, each movie review video was also annotated using five-scale sentiment labels: strongly positive, weakly positive, neutral/ambivalent, weakly negative, and strongly negative; where two raters were involved in annotating the YouTube

Table 2 Scope of our paper compared to other related review papers (where T: Text, A: Audio, V: Visual, P: Physiological, ER: Emotion Recognition, SA: Sentiment Analysis, MT: Medical Treatment, PA: Pain Assessment, SQ: Sleep Quality, and CA: Cognition Analysis)

Ref.	Period	# Datasets	Systematic	Comprehensive	Challenges	Ethics	Modalities					Applications				
							T	A	V	P	ER	SA	MT	PA	SQ	CA
(Poria et al. 2017)	2000-2013	8		✓			✓	✓	✓		✓					✓
(Swain et al. 2018)	1996-2017	59						✓			✓					✓
(Saxena et al. 2020)	2002-2017	14					✓	✓		✓	✓					✓
(Mellouk and Handouzi 2020)	1998-2019	13							✓		✓					✓
(Akçay and Oğuz 2020)	1997-2020	24						✓			✓					✓
(Chandrasekaran et al. 2021)	2019	10		✓			✓	✓			✓					✓
(Kusal et al. 2021)	2005-2020	11			✓		✓				✓					✓
(Wang et al. 2022)	2001-2021	7	✓				✓	✓		✓	✓					✓
(Zhou et al. 2022)	1997-2020	19						✓			✓					✓
(Jemioło et al. 2022)	2008-2019	18								✓	✓					✓
(Siddiqui et al. 2022)	2021	39		✓			✓	✓		✓	✓					✓
(Tawsif et al. 2022)	2009-2021	11								✓	✓					✓
(Atmaja et al. 2022)	2002-2022	19	✓				✓	✓			✓					✓
(Zhu et al. 2023)	2011-2020	9			✓		✓	✓			✓					✓
Ours	2000-2023	63		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

videos and one rater was involved in annotating the ExpoTV videos taken into consideration the original evaluation.

MOUD (Multimodal Opinion Utterances Dataset) (Pérez-Rosas et al. 2013a): This is a multimodal dataset composed of 80 videos obtained from YouTube using keywords that are likely to prompt a product review or recommendation. The videos were expressed in Spanish by 15 male and 65 female speakers, with ages approximately ranging from 20 to 60 years old. Introductory titles and advertisements were eliminated from each video. To ensure that each video focuses on a single topic, a manual trimming process was employed to select a 30-second segment containing opinions. The review videos were divided into segments based on utterances, with an average of six utterances per segment, and an average of 5 s and a standard deviation of 1.2 s per utterance. The final dataset is composed of 498 utterances, where each utterance was transcribed, the dataset was manually annotated as 182 positive, 231 negative and 85 neutral labeled utterances using two annotators using the Elan tool with an inter-annotator agreement of 88% and a Kappa of 0.81.

Spanish Multimodal Opinion Dataset (Pérez Rosas et al. 2013a): This is another multimodal dataset for Spanish sentiment analysis created by the same authors of MOUD. It is composed of 105 videos in Spanish collected from YouTube using different keywords: my opinion, my favorite products, I like, I dislike, baby products, my favorite perfumes, recommended movies, political opinion, video games, and animal abuse. The videos were expressed by 21 males and 84 females with ages ranging from approximately 15 to 60 years old from different Spanish-speaking countries (such as Spain, Mexico, and various South American countries). The duration of each video ranges from 2 to 8 min. The same pre-processing of the YouTube dataset (Morency et al. 2011) and MOUD dataset (Pérez-Rosas et al. 2013b) was followed to eliminate introductory titles and segment videos to express a single topic. The videos were manually transcribed and annotated as 47 positive, 54 negative, and 4 neutral using two annotators with average inter-annotator agreement of 92%, and a kappa of 0.84. The authors of this Spanish dataset also constructed an English multimodal dataset of 37 videos collected from ExpoTV. The videos were transcribed using Amazon service and evaluated by one person. The rating information of ExpoTV videos was utilized to annotate the videos such that videos with one or two stars were labeled as negative while videos with four or five stars were labeled as positive. Out of the 37 videos, 20 videos were labeled as positive and 17 videos were labeled as negative reviews. None of these videos had 3 stars and subsequently no neutral class.

News Rover (Ellis et al. 2014): This is a multimodal dataset composed of 929 news videos from different American news programs and channels. These videos were expressed by eight speakers for different roles (politicians, anchors, reporters, pundits/analysts) and recorded between 13th August and 25th December in 2013. They were pre-processed for sentence-length analysis with a duration ranged between 4 and 15 s. The dataset was annotated for three sentiment levels (negative, neutral, positive) by three annotators using Amazon Mechanical Turk.

MOSI (Zadeh et al. 2016): This is a multimodal dataset for sentiment intensity analysis composed of 93 videos in English from YouTube. The videos were expressed using 89 distinct speakers, including 41 female and 48 male speakers, from different ethnic backgrounds. Most of the speakers were approximately between the ages of 20 and 30 years old. The video clips were transcribed manually containing details about pause fillers (such as “umm” and “uhh”), stresses, and speech pauses. The videos were divided into 3,702

segments, which were annotated as 2,199 subjective segments and 1,503 objective ones by two annotators with Krippendorff's alpha of 0.68. The 2,199 subjective segments were then annotated by five annotators categorizing them as strongly positive, positive, weakly positive, neutral, weakly negative, negative, strongly negative, and uncertain. The inter-annotator agreement, measured by Krippendorff's alpha, was found to be 0.77.

AMMD (Arabic MultiModal Dataset) (Alqarafi et al. 2017): This is a visual-textual dataset containing 40 Arabic videos from YouTube. The opinions were expressed by 13 different speakers (three females and ten males) coming from four Arabic-speaking countries covering Modern Standard Arabic (MSA), the Gulf (Khaliiji) dialect, and the Levantine dialect. The videos were manually segmented into 830 utterances, then were transcribed and annotated as 467 objective and 363 subjective utterances by two annotators. The subjective utterances were then manually annotated as positive, negative, or neutral.

Another audio-visual dataset is presented in (Najadat and Abushaqra 2018) containing 21 Arabic videos from the YouTube. A set of search keywords was prepared to retrieve these videos, including: "My opinion", "My experience", and "Public opinion". Thus, the contents of the videos were not related to a particular topic. The videos were manually segmented to the beginning of the opinion utterance and each video was manually annotated as positive, negative or neutral. The duration of each pre-processed video varies from 10 to 40 s.

3.2 Multimodal sentiment and emotion analysis datasets

CMU-MOSEI (CMU Multimodal Opinion Sentiment and Emotion Intensity) Dataset (Zadeh et al. 2018): It is composed of 3228 videos collected from YouTube for 250 topics and were expressed by 1000 speakers (57% males and 43% females) from different ethnic backgrounds. The final set of videos were then tokenized into 23,453 sentences using punctuation markers manually provided by transcripts. Each sentence was annotated for sentiment on a Likert scale $[-3, 3]$ and Ekman emotions of {happiness, sadness, anger, fear, disgust, surprise} using three crowd-sourced judges from the Amazon Mechanical Turk platform.

MELD (Multimodal EmotionLines Dataset) (Poria et al. 2018): It is an extension and enhancement of EmotionLines and contains about 13,700 utterances from 1,433 multi-party conversational dialogues from the TV-series Friends. Each utterance was annotated with Ekman's emotions labels and sentiment labels (negative, positive, and neutral) using three annotators. The overall Fleiss kappa scores of 0.43 and 0.91 were reported for emotion and sentiment annotation, respectively. Another version of MELD was prepared, known as dyadic MELD, from the non-extendable contiguous dyadic sub-dialogues of MELD.

3.3 Multimodal emotion recognition datasets

BELFAST Dataset (Douglas-Cowie et al. 2000): It is an audiovisual dataset containing 239 clips recorded by 100 people. Out of which, 209 clips were collected from the TV programs, including (1) chat shows, (2) religious programs (3) programs tracing the life of real people over time (4) current affairs programs, and 30 clips from the interview recordings. They were annotated as dimensional (activation and evaluation) and categorical affective states.

EmoTV Dataset (Abrilian et al. 2005): It is an audio-visual dataset of 51 video clips of 12 min recorded from French TV channels. It covers interviews of 48 subjects on 24 differ-

ent topics. The interviews were conducted during news segments, involving two individuals, with only one being visible, and there was no spoken feedback from the interviewer.

An audio-visual dataset is presented in (Caridakis et al. 2007) containing 240 samples recorded by subjects from French, German, Greek, Hebrew, and Italian. It contains facial expressions, body movement, gestures and speech. Subjects were asked to act eight emotional states: anger, despair, interest, pleasure, sadness, irritation, joy, and pride, with equal distribution in the valence-arousal space. For each emotion, subjects were asked to perform a specific gesture. Every actor performed each emotion three times.

HUMAINE Dataset (Douglas-Cowie et al. 2007): It contains 50 clips collected from nine natural and induced datasets. They were selected to reflect different emotional dimensions (positive and negative, active and passive) and all the major combinations (consistent emotion, co-existent emotion, emotional transition over time) with varying intensities. The selection encompasses cues from gesture, facial expressions, voice, movement, action, and words, and includes diverse representations of genders and cultures. The selected clips were recorded in different contexts (static, dynamic, indoor, outdoor, monologue, and dialogue) and expressed in different languages: English, German, French, and Hebrew. The labeling scheme was created to capture three levels of description: the perceived emotional content, the signs that convey emotion, and contextual factors aiding in the interpretation of these cues.

IEMOCAP (Interactive Emotional Dyadic Motion Capture Dataset) (Busso et al. 2008): It is composed of twelve hours of data recorded from ten actors (five males and five females) in dyadic sessions (five sessions with two subjects each). Two recording scenarios were followed. For the first scenario, the actors were asked to perform three selected scripts with clear emotional content. For the second scenario, the subjects were also asked to improvise dialogues in hypothetical scenarios, designed to elicit specific emotions. Detailed information about facial expressions and hand movements of the actors was provided. The final dataset is composed of 10,039 turns. Out of them, 5,255 turns were obtained from scripted sessions and 4,784 turns were obtained from spontaneous sessions. The average duration of turns is 4.5 s and the average value of words per turn was 11.4 words. The audio dialogues were transcribed using Ubiquis (now Acolad¹). Two emotional models were considered to annotate IEMOCAP: discrete categorical-based annotations (anger, sadness, happiness, frustration, and neutral state), and continuous attribute-based annotations (i.e., activation, valence, and dominance). The emotional content of each utterance was evaluated using three evaluators. Additionally, the self-assessment manikins (SAMs) were used to evaluate the corpus in terms of the attributes valence, activation, and dominance. Two distinct evaluators also assessed the emotional content of the dataset utilizing the SAMs system.

Vera am Mittag (Grimm et al. 2008): This is another multimodal dataset composed of 112 h of audio-visual recordings from the German TV talk show “Vera am Mittag”. The recordings were segmented into broadcasts, dialogue acts, and utterances. The dataset encompasses spontaneous and highly emotional speech captured during unscripted, genuine discussions among talk show guests. These emotion labels were assigned on a continuous-valued scale of valence, activation, and dominance.

eINTERFACE (Martin et al. 2006): This is an audio-visual emotion dataset containing the six archetypal emotions experienced by 46 human subjects from 14 countries. Six successive short stories were prepared with assigning a particular emotion for each story. Each

¹ <https://www.acolad.com/>

human subject, then, listened to the stories and reacted to each of the situations. The human participant was instructed to read, memorize, and pronounce (one at a time) the five suggested utterances, each representing a distinct reaction to the provided scenario. Participants were encouraged to infuse as much expressiveness as possible, generating messages focused solely on conveying the intended emotion. Two experts were involved to evaluate whether the reaction clearly expressed the requested emotion or failed to express the emotion. If so, the video segment was accepted otherwise the subject was asked to try again. The segment was discarded if the subject could not react the emotion after several attempts. The dataset is composed of 1,166 video sequences (264 from female recordings and 902 from male recordings).

SEMAINE (McKeown et al. 2010): This is an audio-visual dataset created for building agents that can interact with humans in an emotional conversation using a Sensitive Artificial Listener (SAL). Twenty participants were recorded while having a conversation with (human) operator using two distinct setups: Solid SAL, crafted to ensure operators exhibited suitable non-verbal behavior, and Semi-automatic SAL, structured to emulate users interacting with a machine. In addition to their roles in interaction, four participants assumed the position of the operator during the sessions. The operator and the participant were situated in separate rooms, with visual contact facilitated through teleprompter screens and auditory communication via speakers. It was compiled for 100 conversational and 50 non-conversational recordings, each lasted for approximately 5 min. The dataset was fully annotated manually by four annotators on a continuous value scale of five dimensions of valence, activation, power, anticipation/expectation with the addition of overall emotional intensity. The ages of the participants ranged between 22 and 60 years from eight different countries with Caucasian backgrounds, with 62% of them were female.

SEMAINE2 (McKeown et al. 2011): This is an extended version of *SEMAINE* (McKeown et al. 2010) that is composed of 150 participants, resulting in a total of 959 conversations with individual SAL characters, each lasted for approximately 5 min. The recordings underwent transcription and thorough annotation, covering five affective dimensions and 27 associated categories. They obtained from the Solid SAL configuration were annotated by 6–8 annotators, while two annotators evaluated the recording obtained from Semi-Automatic SAL configuration and one annotator for Automatic SAL sessions.

Piim News (Pereira et al. 2019): It consists of 960 news videos acquired from 51 broadcasts of three Brazilian news programs—Jornal da Band, Jornal da Record, and Jornal Nacional—along with an American television news channel (CNN). The videos were segmented ensuring that each video contains only a single news story to prevent identical tension levels. The duration of videos ranged between 3 s to 815 s. Each segment was annotated as either low or high tension by seven human annotators. Out of the dataset, 619 videos were annotated as low tension whereas 314 videos were annotated as high tension.

Telugu Movie Songs (Abburi et al. 2016): This is an audio-textual dataset created for analyzing emotions in Telugu movie songs. It is composed of 300 clips with corresponding lyrics collected from YouTube. It was annotated by three annotators as happy (also represents Joyful, thrilled, powerful, etc) or sad (also represents ignored, depressed, worry, etc).

VideoEmotion (Jiang et al. 2014): This is an audio-visual dataset composed of 1101 user-generated videos collected from YouTube and Flickr. Ten annotators (five males and five females) assigned labels to each video, categorizing them as expressing anger, anticipation, disgust, fear, joy, sadness, surprise, or trust. The average duration of each video is 107 s.

EMOREACT (Nojavanasghari et al. 2016): This is a multimodal audio-visual dataset for analyzing the emotional states of children between the ages of four to fourteen years old. It is composed of 1,102 audio-visual clips gathered from 63 children. EMOREACT was annotated by three independent evaluators for 17 distinct emotional states: six basic emotions, neutral, valence, and nine complex emotions such as curiosity, uncertainty, and frustration.

ElderReact (Ma et al. 2019): It is a multimodal audio-visual dataset collected for analyzing older adults' emotions including anger, disgust, fear, happiness, sadness, and surprise as well as valence. It is composed of 1,323 video clips recorded by 46 subjects.

Another multimodal dataset is presented in (Pereira et al. 2016) containing 520 news videos, from three Brazilian and one American popular TV newscasts. It was created for analyzing the tension levels during news videos. It was annotated manually as low or high tension by four professionals from several fields of knowledge. An agreement rate of 73.27% was reported for annotating 381 videos, with an additional 75% agreement for annotating 96 videos. In addition to the audio and visual modalities, textual data from closed captions were extracted for 264 videos.

An audio-textual dialogue dataset is presented in (Wu and Liang 2011). It consists of 2,033 utterances recorded by eight Chinese users within a controlled laboratory setting. The dataset collection involved two distinct scenarios. The first scenario, Corpus A, involved a dialogue system guiding speakers to discuss their daily lives by posing questions and recording their responses with acted emotional states. In the second scenario, Corpus B, subjects expressed their emotions through interactions with a computer game. The speaker of the utterance was asked to annotate the corresponding sentence manually as angry, happy, neutral, or sad emotional states. Another evaluator was involved to annotate the data.

3.4 Multimodal sarcasm detection datasets

Multimodal Sarcasm Detection Dataset (MUSARD) (Castro et al. 2019): It was created for sarcasm detection and compiled from popular TV shows. A set of search keywords such as "Friends sarcasm", "Chandler sarcasm", "Sarcasm 101", "Sarcasm in TV shows" was used to retrieve videos from the TV shows: Friends, The Golden Girls, Sarcasmaholics Anonymous, and The Big Bang Theory. The dataset was annotated as non-sarcastic and sarcastic using two annotators and a third annotator was involved for reconciling the disagreement. Out of 6,365 videos, 345 videos were labeled as sarcastic and 6,020 videos were labeled as non-sarcastic. Hence, this dataset is highly imbalanced.

3.5 Multimodal sentiment and demographics recognition

Sentiment Analysis Dataset for Arabic Multimodal (SADAM) (Al-Azani and El-Alfy 2020): This is a multimodal dataset that was sourced from user-generated YouTube videos, and annotated for sentiment and demographics including gender, age group, nationalities, and dialect. It comprises a total of 63 opinion videos, expressed by 37 males and 26 females. It was manually segmented into 524 utterances, capturing diverse expressions. Speakers, aged approximately between 15 and 65, shared their opinions across various topics, and from different countries. This dataset encompasses four age groups: Young adults, Middle-aged I, Middle-aged II, and Senior. The opinions cover diverse domains, including product and movie reviews, cultural perspectives, etc. Additionally, the dataset represents four Arabic

dialects: Egyptian, Levantine, Gulf, and Maghrebi. It was recorded in a variety of settings such as homes, studios, offices, cars, or outdoor environments. These user-generated videos span different time periods, providing a rich and varied collection of opinions and demographic information. SADAM (Al-Azani and El-Alfy 2020) is an extended version of an earlier dataset by the same authors, known as *Multimodal Sentiment and Gender Recognition (MSARDS) dataset* (Al-Azani and El-Alfy 2019b).

3.6 Physiological and medical condition monitoring

DREAMER (Katsigiannis and Ramzan 2018): It comprises EEG and ECG signals captured during affect elicitation through audio-visual stimuli. Signals from 23 participants were recorded, alongside their self-assessment of affective states in terms of valence, arousal, and dominance after each stimulus. The data was acquired using portable, wearable, wireless, cost-effective, and readily available equipment, demonstrating the potential for applying affective computing methods in everyday applications.

Multimodal Genuine Emotion and Expression Detection (MGEED) (Wang et al. 2023): This is a multimodal emotional dataset consisting of facial image sequences, depth maps, EEG, optomyography (OMG), and ECG signals collected from 17 participants. MGEED was annotated in both continuous and discrete emotional states including the 9-level valence, arousal, and the 6 basic emotions.

Asian Affective and Emotional State (A2ES) (Ab. Aziz et al. 2023): This is an emotional multimodal dataset that encompasses ECG and photoplethysmography (PPG) recordings obtained from 47 participants of diverse Asian ethnicities. The participants were exposed to 25 meticulously chosen audio-visual stimuli designed to evoke specific targeted emotions. This work includes an examination of the participants' self-assessments and a detailed list of the 25 stimuli was employed in the study.

Bio-Reactions and Faces for Emotion-based Personalization (BIRAFFE) (Kutt et al. 2019): It is a multimodal dataset of ECG, GSR, and changes in facial expression signals. BIRAFFE was collected from 206 participants using audio-visual stimuli and a gaming session. The signals were acquired using affordable and portable equipment, specifically the BITalino (r)evolution kit for ECG and GSR, and the Creative Live! webcam for capturing face photos (subsequently analyzed using the MS Face API).

Bio-Reactions and Faces for Emotion-based Personalization v2 (BIRAFFE2) (Kutt et al. 2022): This is an extended version of (BIRAFFE) (Kutt et al. 2019), composed of accelerometer, ECG and EDA signals, facial expression, and questionnaires on participants' personality and game engagement. BIRAFFE2 was collected using audio-visual emotional stimuli and a gaming session, from 102 participants.

Women and Emotion Multi-modal Affective Computing (WEMAC) (Miranda et al. 2022): It is a multimodal dataset of physiological and audio signals collected from 47 volunteer women to analyze the violence against women.

SenseEmotion (Velana et al. 2017): This dataset includes recordings of skin conductance level, ECG, electromyography (EMG), respiration (RSP), visual facial regions, and audio signals. The data can be used to detect patterns of heat pain intensities influenced by emotional stimuli and was collected from 45 healthy participants.

K-EmoCon (Park et al. 2020): This is a multimodal dataset of videos capturing visual modalities (face, gesture), speech audio, accelerometer, and biosignals (EEG, ECG, BVP,

EDA, skin temp). It was collected 32 human subjects engaged in naturalistic conversations during 16 sessions of approximately 10-minute-long paired debates on a social issue. This dataset was annotated to represent arousal and valence as continuous emotional states, along with 18 categorical emotional states.

ECSMP (Gao et al. 2021): It is a multimodal dataset of EEG, ECG, PPG, EDA, temperature, and accelerometer data, recorded from 89 healthy college students. The recordings encompass various states, including resting state, emotion induction and recovery, and a series of cognitive function assessment tasks. Quantitative assessments of cognitive functions, including sustained attention, response inhibition, working memory, and strategy use, were conducted using the Cambridge Neuro-psychological Test Automated Battery (CAN-TAB). The sleep ECG data, gathered the night preceding the emotion-induction experiment, were analyzed to evaluate sleep quality.

DSDRWD (Healey and Picard 2005): It is a multimodal dataset of video, ECG, EMG, skin conductance, and respiration collected during real-world driving tasks from 24 drives for stress level analysis.

StressID (Chaptoukaev et al. 2023): This is another multimodal dataset for stress identification, which is one of the largest datasets that contains recordings of 65 participants and above 39 h of annotated data of multiple modalities (EDA, ECG, Respiration, Face video, and Speech).

RECOLA (Ringeval et al. 2013): This multimodal dataset of audio, video, ECG, and EDA (GSR) was collected from 46 subjects in dyads during a video conference.

SWELL-KW (Koldijk et al. 2014): A multimodal dataset of visual, body posture, ECG, and EDA (GSR) collected from 25 participants while carrying out typical knowledge work including writing reports, reading e-mails, searching for information, and making presentations. It was recorded under different working conditions with stressors: neutral, interruptions, and time pressure.

WESAD (Schmidt et al. 2018): This is a multimodal dataset of blood volume pulse, ECG, EDA, electromyogram, respiration, body temperature, and three-axis acceleration collected for wearable stress and affect detection. WESAD was collected from 15 participants during a lab study.

4 Critical analysis and trend of surveyed multimodal datasets

4.1 Characteristics analysis

This section outlines a set of features to describe the fundamental characteristics of multimodal datasets and defines a taxonomy of multimodal emotional datasets based on several criteria including modalities (textual, audio, visual, physiological), composition type (homogeneous or heterogeneous), type of data (natural, acted, induced), task (e.g. emotion recognition, sentiment analysis, etc), data source (social media or TV channels), environment of data collection, emotional model, annotation level, and language. This taxonomy is illustrated in Fig. 1 and Table 3 summarizes the datasets based on the set of defined attributes. The taxonomy and analysis provide insights into the strengths and limitations (drawbacks) of each dataset. For example, if a dataset is recorded in controlled setting, it may fail to capture spontaneous emotions. Additionally, a dataset with only two modalities

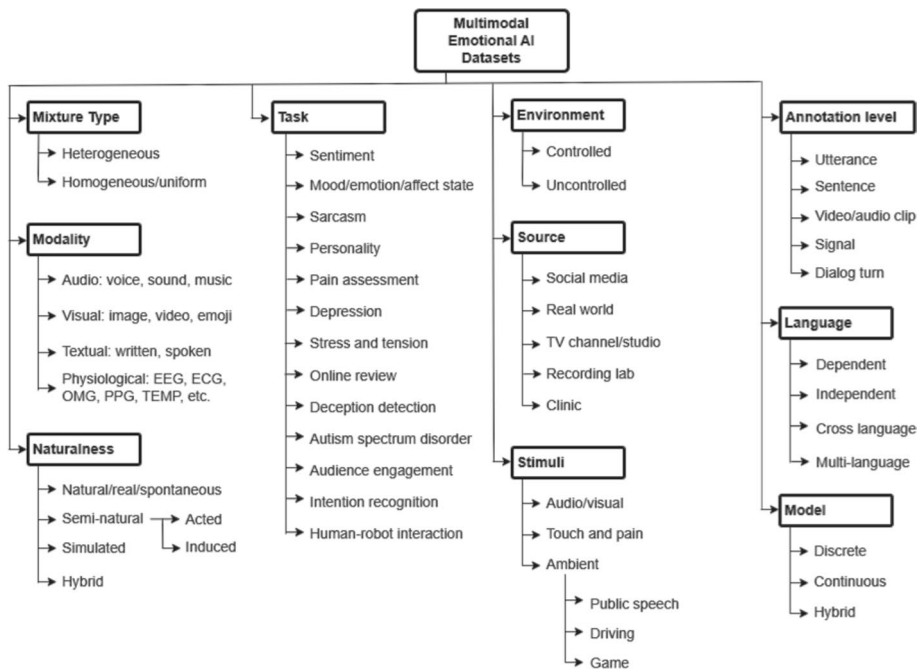


Fig. 1 Taxonomy of emotional AI datasets based on various characteristics and features

may lack the depth needed to represent full range of emotions. A dataset annotated at the utterance level provides greater granularity and fine-grained scale to the one only annotated at the video level.

A) Mixture type: This can be homogeneous or heterogeneous. A homogeneous multimodal dataset consists of multiple types of data modalities that are related to the same topic, context, or task. In other words, the different types of data in the dataset are all closely related and pertain to a common theme. Those datasets are presented in Table 3. On the other hand, a heterogeneous multimodal dataset contains data modalities that are diverse and may not be directly related to each other. These datasets involve different types of data that might not share the same context or task such as a dataset containing social media posts with associated images and audio clips. The retrieved heterogeneous datasets, presented in Table 4.

B) Modality: Some datasets were prepared with more than two modalities including visual, audio, textual, and physiological signals. Others were prepared for bi-modality of audio-visual, audio-textual, and visual-textual. Table 3 presents information on the modalities considered for each reviewed dataset. Textual modality can be obtained from the closed captions such as (Pereira et al. 2016), speech transcriptions such as YouTube (Morency et al. 2011), (Wu and Liang 2011), ICT-MMMO (Wöllmer et al. 2013), MOUD (Pérez-Rosas et al. 2013), SMOD (Pérez Rosas et al. 2013), News Rover (Ellis et al. 2014), MOSI (Zadeh et al. 2016), Telugu (Abburri et al. 2016), MOSEI (Zadeh et al. 2018), MELD (Poria et al. 2018), MMSGDS (Al-Azani and El-Alfy 2019b), Piim News (Pereira et al. 2019), MUSTARD (Castro et al. 2019), SADAM (Al-Azani and El-Alfy 2020), M3ED (Zhao et al. 2022b), or the context such as VideoEmotion (Jiang et al. 2014). The creators of VideoEmo-

Table 3 Homogeneous multimodal datasets. A: Audio, V: Visual, T: Textual, P: Physiological, CE: Categorical Emotion, ARO: Arousal, VAL: Valence, DOM: Dominance, NA: Not Applicable, NM: Not Mentioned, utt.: Utterance

Dataset	Year	Task	Modality	Type of data	Language	Size	No of subjects	Model	Level
Belfast (Douglas-Cowie et al. 2000)	2000	Emotion	A, V	Natural, Acted	English	239 clips	100		
EmoTV1 (Abrilian et al. 2005)	2005	Emotion	A, V	Natural	French	51 videos, 281 utt.; 12 m	48	14 class, 2-dim Intensity and VAL	
DSDRWDT (Healey and Picard 2005)	2005	Stress	V, P	Natural	NA	30 m per driver	24	Stress	
eNTERFACE (Martin et al. 2006)	2006	Emotion	A, V	induced	English	1166 utt	42	6-class	Utterance
HUMAINE (Douglas-Cowie et al. 2007)	2007	Emotion	A, V	Natural + induced	English, German, French, Hebrew	50 clips			
(Caridakis et al. 2007)	2007	Emotion	A, V	Acted		240	10 6 M, 4F	8-class Emotion, 2 VAL, 2 ARO, 8 gestures	
IEMOCAP (Busso et al. 2008)	2008	Emotion	A, V	Acted	English	1039 turns; 12 h	10	5-class discrete, 3-class continuous	Dialog turn level
VAM (Grimm et al. 2008)	2008	Emotion	A, V	Natural	German	1018 utt	47 (11 m, 36f)	3-class dim	
SEMAINE (McKeown et al. 2010)	2010	Emotion	A, V	Induced	English	100 character conversational and 50 non-conversational recordings	20		
YouTube (Morency et al. 2011)	2011	Sentiment	A, V, T	Natural	English	47 videos, 498 utt.; 29 m	47	3-class +, -, =	Utterance
SEMAINE2 (McKeown et al. 2011)	2011	Emotion	A, V	Induced	English		150	5 aff. dim, 27 associated categories	Utterance
- (Wu and Liang 2011)	2011	Emotion	A, T	Induced	Chinese	2,033 utt./sentences; 20 h	8	4-class	
ICT-MMMO (Wöllmer et al. 2013)	2013	Sentiment	A, V, T	Natural	English	370 videos; 14 h	200	3-class +, -, =	Video
MOUD (Pérez-Rosas et al. 2013)	2013	Sentiment	A, V, T	Natural	Spanish	80 Video, 498 utt		3-class	Utterance

Table 3 (continued)

Dataset	Year	Task	Modality	Type of data	Language	Size	No of subjects	Model	Level
SMOD (Pérez Rosas et al. 2013)	2013	Sentiment	A, V, T	Natural	Spanish	105 videos 550 utt		3-class	Video
English videos (Pérez Rosas et al. 2013)	2013	Sentiment	A, V, T	Natural	English	37		2-class	Video
RECOLA (Ringeval et al. 2013)	2013	Emotion	A, V, P	Natural	French	4 h	46: 19 M, 27F	ARO, VAL	
News Rover (Ellis et al. 2014)	2014	Sentiment	A, V, T	Natural	English	929 videos	9	3-class +, -, =	Sentence
VideoEmotion (Jiang et al. 2014)	2014	Emotion	A, V, T	Natural	English	1,101 videos		8-class	
SWELL-KW (Koldijk et al. 2014)	2014	Emotion, stress	V, P	Natural	NA	75 h	25	ARO, VAL, Stress	
DECAF (Abadi et al. 2015)	2015	Emotion	P	Induced	NA		30	2: ARO, VAL	Signal
MOSI (Zadeh et al. 2016)	2016	Sentiment	A, V, T	Natural	English	93 videos, 3702 utt	89	2-class subj./obj., 8-class sent., 4 gestures	Utterance
- (Pereira et al. 2016)	2016	Tension	A, V, T	Natural	English	520 videos		2-class Low, High tension	
Telugu (Abburi et al. 2016)	2016	Emotion	A, T	Natural	Telugu (Indian)	300 songs	NM	Happy, Sad	Whole, begin/end segments
ASCERTAIN (Subramanian et al. 2016)	2016	Emotion	P	Induced	NA	36 stimuli	58	2: ARO, VAL	
MSP-IMPROV (Busso et al. 2016)	2016	emotion	A, V	Acted	English		12	11: Emotions	Utterance
EMOREACT (Nojavanasghari et al. 2016)	2016	Emotion	A, V	Natural	English	1102	63: 31 M, 32F	17 CE	
AMMD (Alqarafi et al. 2017)	2017	Sentiment	V, T	Natural	Arabic	40 Videos, 830 utt., 100 m	13 (10 M, 3F)	2-class polarity, 2-class subj	Utterance
SenseEmotion (Velana et al. 2017)	2017	Pain assessment	A, V, P	Induced		23 h	45	3 pain levels	
MOSEI (Zadeh et al. 2018)	2018	Sentiment, Emotion	A, V, T	Natural	English	3228 videos, 23453 utt., 57 h	1000	5-sent., 6-emotions	Utterance
MELD (Poria et al. 2018)	2018	Sentiment, Emotion	A, V, T	Natural	English	1433 dialogues, 13000 utt		3-sentiment, 7-emotion	Utterance
(Najadat and Abushaqra 2018)	2018	Sentiment	A, V	Natural	Arabic	21 videos	-	3-class	

Table 3 (continued)

Dataset	Year	Task	Modality	Type of data	Language	Size	No of subjects	Model	Level
DREAMER (Katsigiannis and Ramzan 2018)	2018	Emotion	P	Induced	NA		25: 14 M, 11 F	3: ARO, VAL, DOM	Signal
AMIGOS (Miranda-Correa et al. 2018)	2018	Emotion	P	Induced	NA	24 stimu	40	3: ARO, VAL, DOM	
WESAD (Schmidt et al. 2018)	2018	Stress, amusement	P	Induced, Natural	NA	10stimuli	15	Neutral, stress, amusement	
MMSGDS (Al-Azani and El-Alfy 2019b)	2019	Sentiment, gender	A, V, T	Natural	Arabic	63 videos, 524 utt		4-class	Utterance
Pim News (Pereira et al. 2019)	2019	Emotion	A, V, T	Natural	English	960 Videos		2-class (low, high) tension	
MUSTARD (Castro et al. 2019)	2019	Sarcasm	A, V, T	Natural	English	6,365 videos		2-class sarcastic, non-sarcastic	Utterance
BIRAFFE (Kutt et al. 2019)	2019	Personality	V, P	Induced			141	5-personality traits	
ElderReact (Ma et al. 2019)	2019	Emotion	A, V			1323 video clips	46: 20 M, 26F	6-basic emotions+ VAL	
SADAM (Al-Azani and El-Alfy 2020)	2020	Sentiment, Demog	A, V, T	Natural	Arabic	63 videos, 524 utt		2-class +, -, 2 gender, 4 age-group, 4 dialect	
K-EmoCon (Park et al. 2020)	2020	Emotion	A, V, P	Natural	English	173 minutes	32	2: ARO, VAL, 18 CE	
ECSMP (Gao et al. 2021)	2021	Emotion, cognition, sleep	P	Induced, Natural	NA	6: stimuli	89	6: CE, 12 sleep indices	
WEMAC (Miranda et al. 2022)	2022	Emotion	A, P	Induced	NA	17 hours phys. Signals, 6 hours audio signals	47: women	VAL, ARO, 12 emotions	Signal
M3ED (Zhao et al. 2022b)	2022	Emotion	A, V, T	Natural	Chinese	9,082 turns and 24,449 utt		7-class	Utterance
BIRAFFE2 (Kutt et al. 2022)	2022	Personality	V, P, Q	Induced	NA		102	5-personality traits	
MGFEED (Wang et al. 2023)	2023	Emotion	V, P	Induced	NA	150k images	17	ARO, VAL, 6 basic emotions	Frame
A2ES (Ab. Aziz et al. 2023)	2023	Emotion	P	Induced	NA		47		

Table 3 (continued)

Dataset	Year	Task	Modality	Type of data	Language	Size	No of subjects	Model	Level
StressID (Chaptoukaev et al. 2023)	2023	Emotion, Stress	A, V, P	Induced		39 h	65	ARO, VAL, Stress	

Table 4 Characteristics of multimodal heterogeneous datasets

Dataset	Year	Task	Modality	Source/Environment	Language	Total	Model
SentiBank Twitter (Borth et al. 2013)	2013	Sentiment	V, T	Twitter	English	470 +ve, 133 -ve	2-class
Sina Weibo (Wang et al. 2014b)	2014	Sentiment	V, T	Micoblog Sina-Weibo	Chinese	5859 texts with images	3-class
Getty (You et al. 2016)	2016	Sentiment	V, T	Getty Images	English	588,221 images	2-class
– (You et al. 2016)	2016	Sentiment	V, T	Twitter	English	31,584 tweets with images	2-class
– (You et al. 2015)	2015	Sentiment	V, T	Twitter	English	769 +ve, 500 -ve=1967	2-class
– (Al-Azani and El-Alfy 2018)	2018	Sentiment	V, T	Twitter	Arabic		2-class
Tumblr (Hu and Flaxman 2018)	2018	Emotion	V, T	Tumblr/Posts	English	256,897	15- class
Uzbek (Rab- bimov et al. 2020)	2020	Sentiment	Emoji, T	Youtube Comments	Uzbek	2,563	2-class
– (Al-Azani and El-Alfy 2021)	2021	Sentiment	Emoji, T	Twitter, Yout	Arabic	2021	2-class
Multi- ZOL (Xu et al. 2019b)	2019	Sentiment	V, T	ZOL			
Memo- tion (Sharma et al. 2020)	2020	Sentiment	V, T	Facebook, Reddit, etc			3-class
Twitter-15 (Yu and Jiang 2019)	2019	Sentiment	V, T	Twitter			
Twitter-17 (Yu and Jiang 2019)	2019	Sentiment	V, T	Twitter			
MSA- IR (Chen et al. 2023)	2023	Sentiment	V, T	Twitter	English	4600 (1404+ve, 1388 neut., 1808-ve)	3-class

tion dataset (Jiang et al. 2014) noted that certain emotions demonstrate strong correlations with specific semantics, such as the presence of particular events or objects. For instance, the emotion of ‘joy’ might commonly coincide with events like parties and children playing. This insight inspired them to suggest incorporating semantic attributes into video emotion analysis. Visual modality can be also represented as a video (sequence of frames) or single images. In (Caridakis et al. 2007; Park et al. 2020), the visual modality is represented by faces and body gestures. Physiological data plays a crucial role in understanding emotional

states, offering objective measures (Ruotsalo et al. 2024). Various physiological channels, such as heart rate, skin conductance, and facial electromyography, provide valuable insights into emotional responses (Bastida et al. 2024). Additionally, a range of devices is utilized for data collection, including wearable devices like smartwatches and fitness trackers, as well as contact electrodes that measure muscle activity and skin responses. These tools enable continuous and precise monitoring of physiological signals, enhancing our ability to analyze and interpret emotional experiences. Several examples of physiological multimodal datasets are shown in Table 5.

C) Type of data: Multimodal datasets in this study also classified as natural, acted, or induced datasets. Natural datasets represent genuine emotions that naturally occur in real-

Table 5 Various modalities for physiological-based emotional datasets

Dataset	Task	Modality
DREAMER (Katsigiannis and Ramzan 2018)	Emotion	EEG, ECG
DECAF (Abadi et al. 2015)	Emotion	MEG, NIR, hEOG, ECG, tEMG
MGEED (Wang et al. 2023)	Emotion	Image, OMG, EEG, ECG
A2ES (Ab. Aziz et al. 2023)	Emotion	ECG, PPG
WEMAC (Miranda et al. 2022)	Emotion	BVP, GSR, SKT, Audio
BIRAFFE (Kutt et al. 2019)	Personality	ECG, GSR, facial expression
BIRAFFE2 (Kutt et al. 2022)	Personality	accelerometer, ECG and EDA, facial expression, questionnaires
SenseEmotion (Velana et al. 2017)	Pain assessment	biopotentials, visual, audio
K-EmoCon (Park et al. 2020)	Emotion	Videos (face, gesture), speech audio, accelerometer, EEG, ECG, BVP, EDA, skin temp
AMIGOS (Miranda-Correa et al. 2018)	Emotion	ECG, EEG, GSR
ASCERTAIN (Subramanian et al. 2016)	Emotion	ECG, EEG, GSR
ECSMP (Gao et al. 2021)	Emotion, cognition, sleep quality	ECG, EEG, PPG, GSR, ACC, TEMP
DSDRWDT (Healey and Picard 2005)	Stress	Video, ECG, EMG, GSR, RESP
RECOLA (Ringeval et al. 2013)	Emotion	Audio, Visual, ECG, EDA
SWELL-KW (Koldijk et al. 2014)	Emotion, stress	Visual, body posture, ECG, EDA
WESAD (Schmidt et al. 2018)	Stress, amusement	ECG, PPG, GSR, EMG, TEMP, RESP (Respiration)
StressID (Chaptoukaev et al. 2023)	Emotion, stress	EDA, ECG, RESP, Face video, speech
AMHUSE (Boccignone et al. 2017)	Emotion, amusement	Video, depth sequence, EDA, TEMP, blood volume pulse

life situations arising organically in response to genuine events or interactions without any external influence or direction. The acted datasets involve subjects/actors intentionally expressing predefined emotions based on specific instructions or scripts provided by researchers. Actors or participants follow a predetermined scenario or set of guidelines. Induced datasets involve eliciting emotions through external stimuli, such as images, videos, or stories. Participants may respond emotionally to stimuli that are designed to evoke specific emotional reactions such as eNTERFACE (Martin et al. 2006), SEMAINE (McKeown et al. 2010), SEMAINE2 (McKeown et al. 2011), (Wu and Liang 2011), DREAMER (Katsigiannis and Ramzan 2018), DECAF (Abadi et al. 2015), MGEED (Wang et al. 2023), A2ES (Ab. Aziz et al. 2023), WEMAC (Miranda et al. 2022), BIRAFFE (Kutt et al. 2019), BIRAFFE2 (Kutt et al. 2022), SenseEmotion (Velana et al. 2017), AMIGOS (Miranda-Correa et al. 2018), ASCERTAIN (Subramanian et al. 2016). Some datasets can be considered as natural and acted such as Belfast (Douglas-Cowie et al. 2000) and natural and induced such as HUMAINE (Douglas-Cowie et al. 2007). The type of data per each reviewed dataset is presented in Table 3.

D) Task: This attribute defines the purpose or application of building the dataset. Datasets were created for different tasks. Sentiment analysis includes subjectivity classification (Zadeh et al. 2016; Alqarafi et al. 2017), polarity detection (Morency et al. 2011; Wöllmer et al. 2013; Pérez-Rosas et al. 2013b; Pérez Rosas et al. 2013a; Zadeh et al. 2016; Najadat and Abushaqra 2018; Alqarafi et al. 2017), sentiment intensity (Zadeh et al. 2016), sarcasm detection (Castro et al. 2019). Another application is emotion recognition including categorical emotion recognition, dimensional emotion recognition, tension level analysis such as (Martin et al. 2006; Busso et al. 2008; McKeown et al. 2010, 2011; Pereira et al. 2016, 2019; Wu and Liang 2011; Abburi et al. 2016; Jiang et al. 2014; Baveye et al. 2015; Zadeh et al. 2018; Poria et al. 2018). Other applications include stress analysis was considered in DSDRWDT (Healey and Picard 2005), pain assessment in SenseEmotion (Velana et al. 2017), and Personality analytics in BIRAFFE (Kutt et al. 2019). Some datasets were prepared for multiple tasks such as polarity detection and emotion recognition (Zadeh et al. 2018; Poria et al. 2018), subjectivity classification and polarity detection (Zadeh et al. 2016; Alqarafi et al. 2017). A multimodal dataset for sentiment analysis and demographic analysis including gender, age, dialect, and nationality was prepared by (Al-Azani and El-Alfy 2020). ECSMP (Gao et al. 2021) dataset was prepared for Emotion, cognition, and sleep quality while WESAD (Schmidt et al. 2018) was prepared for stress and amusement analysis.

E) Source and environment: This attribute characterizes the origin of the datasets, indicating whether they were recorded in real-world settings, specific environments (such as laboratories), TV studios, or clinical settings. Real-world environments might include different sources such as social media platforms for collecting natural videos such as “YouTube” MOUD (Pérez-Rosas et al. 2013b), SMOD (Pérez Rosas et al. 2013a), MOSI (Zadeh et al. 2016), Telugu (Abburi et al. 2016), AMMD (Alqarafi et al. 2017), MOSEI (Zadeh et al. 2018), (Najadat and Abushaqra 2018), MMSGDS (Al-Azani and El-Alfy 2019b), MUSTARD (Castro et al. 2019), SADAM (Al-Azani and El-Alfy 2020), from ExpoTV such as (Wöllmer et al. 2013; Pérez Rosas et al. 2013a), and from Flickr such as (Jiang et al. 2014). TV channels constitute another valuable source for procuring natural datasets, providing rich and diverse content reflective of real-world scenarios and human expressions. TV channels are considered as a main source for different multimodal datasets such as

EmoTV1 (Abrilian et al. 2005), HUMAINE (Douglas-Cowie et al. 2007), VAM (Grimm et al. 2008), News Rover (Ellis et al. 2014), (Pereira et al. 2016), MELD (Poria et al. 2018), Piim News (Pereira et al. 2019), M3ED (Zhao et al. 2022b). On the other hand studio recording is the main source for recording acted and induced datasets.

Although movies, TV programs, and songs are often recorded in somehow controlled environments, they provide significantly closer to real-world environments than datasets recorded in labs (Dhall et al. 2012). In (D'mello and Kory 2015), the dialogue dataset generated using Wizard-of-Oz (Petrik 2004; Bohus and Rudnicky 2008) experiment is categorized as induced. However, such experiments are expensive and time-consuming to carry out (Serban et al. 2018). Therefore, multimodal datasets can be also classified, in terms of the environment as either lab, close to real-world, or real. Real environmental videos are user-generated, close to real-world environment includes TV and movie recordings. Laboratory environments have the potential to generate a significantly greater volume of raw and controlled data when compared to data generated by users.

F) Genre and topic: Genre categorizes data based on its distinctive style, form, or content characteristics such as action, drama, comedy, and science fiction in the case of movies. Topic categorization involves organizing data based on the subject or theme it addresses such as politics, technology, sports, health, and reviews. The genre of emotional datasets includes reviews, opinions, recommendations, conversational dialogue, songs, and news programs. The datasets can be classified as monologue, dyadic conversational dialogue, and multi-party conversational dialogue in terms of the number of subjects. Monologue datasets contain a single speaker while dyadic conversational datasets contain two speakers. On the other hand, a multi-party emotional conversational dataset contains more than two speakers per dialogue. Sentiment analysis datasets are monologue while emotional datasets are either monologue, dyadic conversational dialogue, or multi-party conversational dialogue. The content of the dataset might belong to a certain topic or cover several topics such as politics, product reviews, religious views, etc. For example, MOSI dataset concentrated on movie reviews (Zadeh et al. 2016) whereas several topics are covered in (Pérez-Rosas et al. 2013a). This criterion is established according to the nature of the study and its intended application. Table 6 presents the genre and topics of the reviewed datasets. This is in addition to the task and the source or/and environment.

G) Language: Some of the datasets might be language-dependent, especially for multimodal sentiment analysis datasets considering textual modality. Each language has its phonemes, morphemes, lexemes, syntax, context, grammar, and semantics. Some languages might have similar components while others are different. As can be seen in Table 3, several datasets were built for English (Morency et al. 2011; Martin et al. 2006; Pereira et al. 2016; Zadeh et al. 2016, 2018; Busso et al. 2008). Other languages are considered including Arabic in (Al-Azani and El-Alfy 2020), Chinese in (Wu and Liang 2011), M3ED (Zhao et al. 2022b), French in RECOLA (Ringeval et al. 2013), German in VAM (Grimm et al. 2008), Indian, and Spanish in MOUD (Pérez-Rosas et al. 2013b) and SMOD (Rosas et al. 2013). HUMAINE (Douglas-Cowie et al. 2007) considered different languages: English, German, French, and Hebrew.

H) Emotional model:

Different methods exist for comprehending and evaluating emotions. The two most commonly employed approaches in affective computing research are (Zachar and Ellis 2012): categorical and dimensional, where the former, led by Ekman (1992), is composed of a

Table 6 Genre and topic of multimodal emotional datasets (CTR stands for ‘close to real-world’)

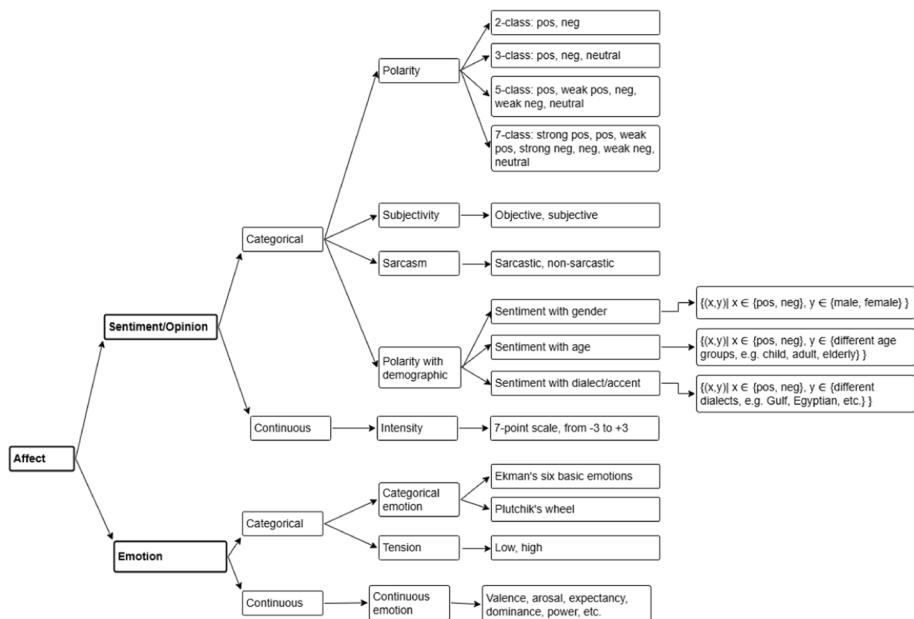
Dataset	Task	Source/Environment	Topic/Scenarios	Genre/Stimuli
Belfast (Douglas-Cowie et al. 2000)	Emotion	TV, Lab	Various	Interviews, TV programs
EmoTV1 (Abrilian et al. 2005)	Emotion	French TV channels	Various	TV interviews
DSDRWDT (Healey and Picard 2005)	Stress	Real: driving within the rest, city, and highway conditions	Specific: driving	Driving task
eNTERFACE (Martin et al. 2006)	Emotion	Lab	Various	Reactions after listening stories
HUMAINE (Douglas-Cowie et al. 2007)	Emotion	TV	Various	Interviews, TV program, dialogs
(Caridakis et al. 2007)	Emotion	Lab		
IEMOCAP (Busso et al. 2008)	Emotion	Lab		Dialogues/dyadic sessions
VAM (Grimm et al. 2008)	Emotion	German TV channel	Various	Personal issues
SEMAINE (McKeown et al. 2010)	Emotion	Lab		interactions between users and an operator
YouTube (Morency et al. 2011)	Sentiment	YouTube	Various	Product reviews, Opinions
SEMAINE2 (McKeown et al. 2011)	Emotion	Lab		Human -Computer Interaction
- (Wu and Liang 2011)	Emotion	Lab		Daily lives and Interactions with Game
ICT-MMMO (Wöllmer et al. 2013)	Sentiment	YouTube, ExpoTV	Specific	Movi reviews
MOUD (Pérez-Rosas et al. 2013b)	Sentiment	YouTube	Various	Product reviews or recommendation
SMOD (Pérez Rosas et al. 2013a)	Sentiment	YouTube	Various	Product reviews, Opinions
English Videos (Pérez Rosas et al. 2013a)	Sentiment	ExpoTV	Specific	Product reviews
RECOLA (Ringeval et al. 2013)	Emotion	CTR: Lab	Various	Dialogues/dyadic sessions
News Rover (Ellis et al. 2014)	Sentiment	TV newscasts	Various	News video
VideoEmotion (Jiang et al. 2014)	Emotion	Flickr, YouTube	Various	
SWELL-KW (Koldijk et al. 2014)	Emotion, stress	CTR: Lab	Specific: knowledge work	Different working conditions with stressors: neutral, interruptions and time pressure
DECAF (Abadi et al. 2015)	Emotion	Lab	Various	Audio-visual stimuli
MOSI (Zadeh et al. 2016)	Sentiment	YouTube	Various	Reviews, opinions
- (Pereira et al. 2016)	Tension	TV newscasts	Specific	News programs

Table 6 (continued)

Dataset	Task	Source/Environment	Topic/Scenarios	Genre/Stimuli
Telugu (Abburi et al. 2016)	Emotion	YouTube	Specific	Movie songs
ASCERTAIN (Subramanian et al. 2016)	Emotion	CTR: Lab	Various	Audio-visual stimuli
MSP-IMPROV (Busso et al. 2016)	emotion	Studio recording	Various	Daydic
EMOREACT (Nojavansghari et al. 2016)	Emotion	YouTube: React channel	Various topics specific for children	Children reacting to 37 subjects
AMMD (Alqarafi et al. 2017)	Sentiment	YouTube	Specific	Product Reviews
SenseEmotion (Velana et al. 2017)	Pain assessment		specific application	Heat stimulation, visual-audio stimuli
MOSEI (Zadeh et al. 2018)	Sentiment + Emotion	YouTube	Various	Reviews, opinions
MELD (Poria et al. 2018)	Sentiment + Emotion	Friends TV series	Various	multi-party conversations episodes
- (Najadat and Abushaqra 2018)	Sentiment	YouTube	Various	Product reviews, Opinions
DREAMER (Katsigiannis and Ramzan 2018)	Emotion	Lab	Various	Audio-visual stimuli
AMIGOS (Miranda-Correa et al. 2018)	Emotion	CTR: Lab	Various	Audio-visual stimuli
WESAD (Schmidt et al. 2018)	Stress, amusement	CTR: Lab	Various	Audio-visual stimuli, cognitive testing
MMSGDS (Al-Azani and El-Alfy 2019b)	Sentiment +gender	YouTube	Various	Reviews, opinions
Pim News (Pereira et al. 2019)	Emotion	TV newscasts	Not specific	News programs
MUSARD (Castro et al. 2019)	Sarcasm	YouTube	Various TV shows	Dialogues
BIRAFFE (Kutt et al. 2019)	Personality	Lab	Personality, Emotion	Audio-visual, gaming session
ElderReact (Ma et al. 2019)	Emotion		Specific for elders	
SADAM (Al-Azani and El-Alfy 2020)	Sentiment + Demographics	YouTube	Various	Reviews, opinions
K-EmoCon (Park et al. 2020)	Emotion	Studio recording	Specific: the issue of accepting Yemeni refugees	Dyadic sessions
ECSMP (Gao et al. 2021)	Emotion, cognition, sleep quality	CTR: Lab	Various	Audio-visual stimuli, cognitive testing
WEMAC (Miranda et al. 2022)	Emotion		Specific for women	28:Audio-visual stimuli
M3ED (Zhao et al. 2022b)	Emotion	TV series	Various	Dialogues/dyadic sessions
BIRAFFE2 (Kutt et al. 2022)	Personality	Lab	Various	Audio-visual, gaming session

Table 6 (continued)

Dataset	Task	Source/Environment	Topic/Scenarios	Genre/Stimuli
MGEED (Wang et al. 2023)	Emotion	Lab	Various	Audio-visual stimuli
A2ES (Ab. Aziz et al. 2023)	Emotion	Lab	Various	Audio-visual stimuli
StressID (Chaptoukaev et al. 2023)	Emotion, Stress	CTR: Lab	Various	Cognitive load, Public speaking, Audio-visual stimuli

**Fig. 2** Affect labels considered in multimodal datasets

countable set of distinct states whereas the later, led by Russell (1980), is composed of two continuous dimensions. Figure 2 provides a taxonomy for affective states in terms of emotion and sentiment for both categorical and continuous based approaches.

For subjectivity classification, samples were annotated as subjective or objective ((Zadeh et al. 2016; Alqarafi et al. 2017)). The scale of polarity classification task includes: binary scale (positive or negative) such as (Wöllmer et al. 2013; Al-Azani and El-Alfy 2020; Pérez Rosas et al. 2013a), 3-point scale (positive, neutral, or negative) such as (Morency et al. 2011; Pérez-Rosas et al. 2013a; Pérez Rosas et al. 2013b; Alqarafi et al. 2017; Najadat and Abushaqra 2018), 5-point scale (strongly positive, positive, neutral, negative, or strongly negative) such as MOSEI (Zadeh et al. 2018), or 7-point scale (strongly positive, positive, weakly positive, neutral, weakly negative, negative, strongly negative, or uncertain).

On the other hand, emotion primitives can be annotated as

- Categorical emotion labels, delineated within a discrete-valued scale, encompass a diverse array of emotional states, with notable examples such as happiness, anger, and sadness. This framework of emotions draws inspiration from renowned models, including Ekman's foundational six universal emotions - Joy, Sadness, Fear, Anger, Surprise, and Disgust which was considered for most of the reviewed datasets such as (Martin et al. 2006; Zadeh et al. 2018). Additionally, Plutchik's influential wheel of emotions (Plutchik 1980) further expands this framework, introducing a comprehensive set of emotions, such as Anger, Anticipation, Disgust, Fear, Joy, Sadness, Surprise, and Trust. Plutchik's model not only presents these emotions but also illustrates their intricate relationships, providing a nuanced understanding of how emotions can manifest and interact in the human experience. This combined framework serves as a valuable tool in the exploration and analysis of the intricate landscape of human emotions. Ekman emotions and Neutral emotion were considered also in (Poria et al. 2018) while facial gestures such as smile, frown, head nod, and head shake were considered in (Zadeh et al. 2016; Alqarafi et al. 2017).
- Continuous dimensional emotion labels are represented on a continuous-valued or real-valued scale, encompassing crucial dimensions such as: valence [-1, 1] which reflects the negative-to-positive spectrum of emotions, arousal [-1, 1] which indicates the intensity or activation level of emotional experiences, expectancy [-1, 1] which signifies the anticipation or predictability associated with emotions, dominance [-1,1] which represents the sense of control or influence felt during emotional states, and power [0,∞].
- Tension: Low or high.
- Stress: Two kind of stress ratings have been considered in DSDRWDT (Healey and Picard 2005): a free scale and a forced ranking of events. In the free-scale section, drivers rated driving events on a scale from "1" to "5" where "1" indicates "no stress" and "5" indicates "high stress". In the forced-scale section, drivers ranked events on a scale from "1" to "7" by assigning "1" to the least stressful event and "7" to the most stressful event.
- Pain: 3 pain levels have been considered in SenseEmotion (Velana et al. 2017).

4.2 Major trends

Figure 3 depicts the distribution of the emotional AI applications and tasks. The age of multimodal affective computing can be divided into three periods. Before 2000, we found that the main focus was on unimodal emotion computing which confirms the finding of (Poria et al. 2017). From 2000 to 2011, there was a direction to apply two modalities, specifically audio and visual modalities for emotion detection and stress analysis. From 2011 to 2023, the focus was shifted on more than two modalities for different applications. From 2019 to 2023, more challenge applications appeared including, Dialogue, Sarcasm, and personality. This is in addition to developing multimodal datasets for multiple objectives or targets.

Additionally, the advances of wearable sensing and related technologies have led to a growing interest in the acquisition and analysis of multi-sensory emotional data captured from physiological signals (Chen et al. 2020). Various physiological signals are employed in emotional recognition systems. Existing literature indicates that emotion recognition systems based on physiological signals offer enhanced robustness in circumventing social masking (Mand et al. 2017). Furthermore, they demonstrate decreased vulnerability to

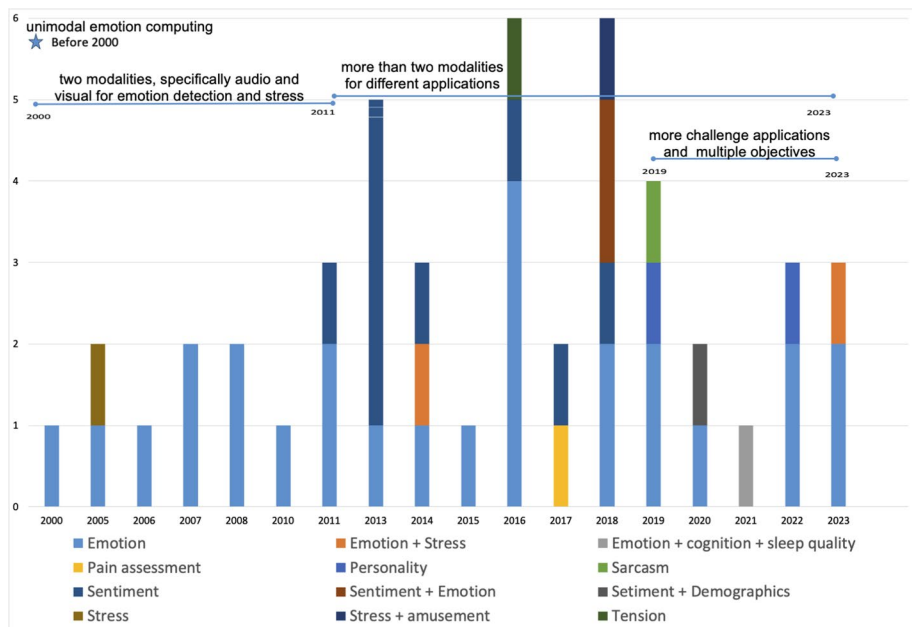


Fig. 3 Distribution of tasks per year for multimodal research from year 2000 to 2023

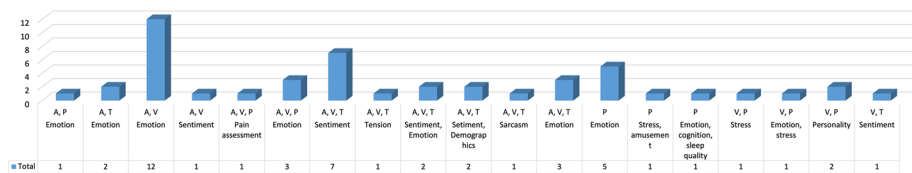
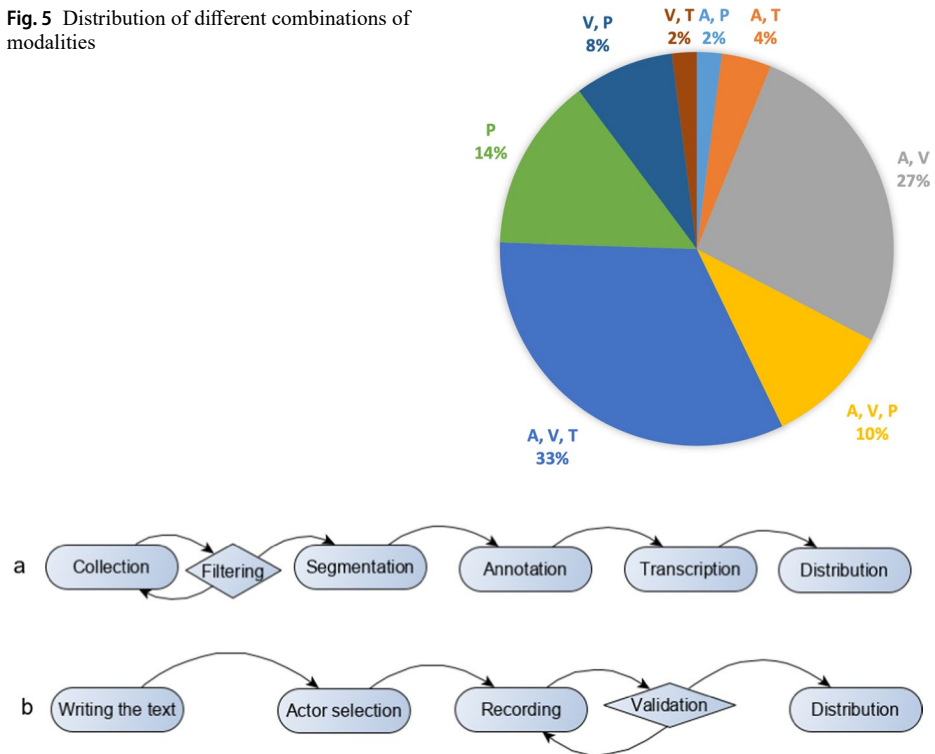


Fig. 4 Distribution of modalities per applications

deceptive expressions and inaccurate emotional states (Hasnul et al. 2021b). However, they are more accurate when combined with non-physiological modalities such as facial expressions (Wang et al. 2023) and audio such as WEMAC (Miranda et al. 2022) or both audio and visual modalities such as K-EmoCon (Park et al. 2020). There is a discernible trend toward the creation of natural datasets, and a noteworthy observation is that starting from 2013, all datasets exhibit a natural or close to real-world characteristic.

Figure 4 depicts the distribution of the considered modalities per task. It can be noticed that audio and visual modalities are the most considered ones for emotion detection, which is followed by Audio, visual, and textual modalities with sentiment analysis. In general, and regardless of the task, audio, textual, and visual are the highest considered modalities representing 34% which are followed by audio, and visual modalities representing 27%. Figure 5 depicts the distribution of modalities. The prevalence of certain modalities than others can be attributed to various factors including their application domain, e.g. text mainly comes with sentiment and rarely comes with emotion, similarly, the physiological signals mainly come with emotion and stress while not with the sentiment. It is also related how humans

Fig. 5 Distribution of different combinations of modalities**Fig. 6** Preparation process for multimodal datasets: a) user-generated b) controlled

commonly interpret emotions on daily basis (e.g. facial expressions and/or focal tone are more common). Other factors may include complexity, the need for specialized sensors, ease and cost of data collection, scalable deployment, etc. For example, while physiological signals can offer valuable insights, they are can be more complex, costly, less scalable, require specialized sensors, etc.

5 Multimodal dataset development: framework, and key requirements and challenges

5.1 Framework

The process of constructing multimodal emotional AI datasets requires a structured framework to ensure data quality, integrity, reliability, diversity, fairness, ethical compliance, and usability (Kye et al. 2017). Numerous intricate steps are required across different modalities in this framework. A typical example is shown in Fig. 6 and its implementation for preparing multimodal datasets from audio, visual and textual data in social media is outlined in Fig. 7. However, the specific steps may vary depending on factors such as the dataset type, the application, and the modalities under consideration. For instance, the procedures for building acted datasets may differ slightly from those for natural spontaneous datasets. Similarly,

the development process for a multimodal dataset involving audio and visual modalities may differ from the one that also includes physiological signals. Despite these variations, several common steps in developing the multimodal emotional datasets surveyed above.

In this context, we outline the shared key steps for developing such datasets. The process begins with planning and defining its purpose, target emotional framework (such as Ekman's emotions or Valence-Arousal-Dominance), and intended applications (such as emotion recognition or sentiment analysis). Next, it focuses on selecting the modalities and data sources to determine the type of data to be collected (e.g. visual, audio, text, physiological). After that, it defines the collection environment whether it will be recorded in controlled lab for induced or acted emotions or through naturalistic recordings to capture spontaneous emotions. Next stage defines how emotions will be elicited, e.g. through controlled stimuli (such as videos, games or virtual reality) or through passive observation in natural interactions (Zafeiriou et al. 2015). To ensure integrity, the process defines and establishes robust collection and synchronization procedures to mitigate environment inconsistencies and sensor noise. For regulatory and ethical considerations, the process includes obtaining informed consent, implementing privacy protection measures, and ensuring compliance with legal regulations such as GDPR or HIPAA. Finally, the process sets up the appropriate sensors (cameras, microphones, wearable, social media, etc.) and identify the participating subjects ensuring demographic diversity to mitigate bias and prevent discrimination in AI models (Adewumi et al. 2024).

Following data collection, the following stages in developing the multimodal emotional dataset include preprocessing steps by filtering and correcting errors, integrating and aligning multiple sources and modalities, splitting the stream into emotional chunks, transforming audio to text (transcription), annotating each chunk to serve as ground truth labels, and distributing and sharing the dataset with detailed documentation. The following two subsections provide more details on the key steps in this framework while describing key requirements and challenges hindering the widespread deployment of these datasets.

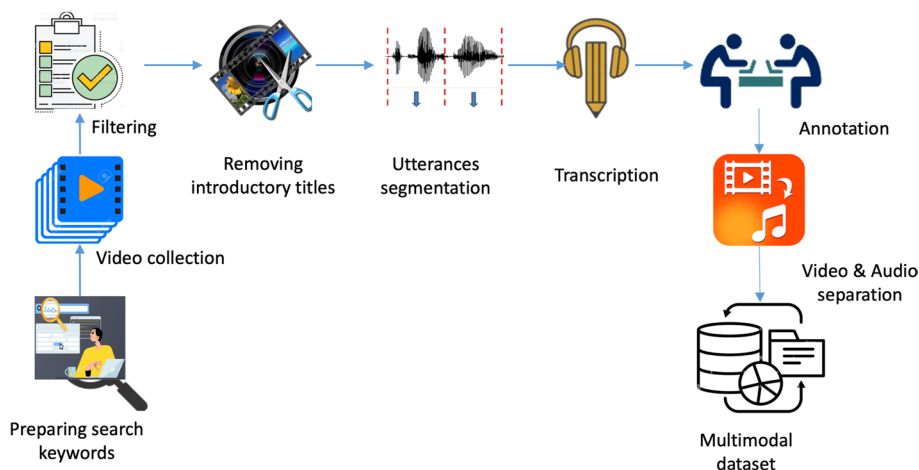


Fig. 7 Preparation process for audio, visual, and textual multimodal datasets collected from social media

5.2 Key requirements

This section discusses the key criteria necessary for successful development of multimodal emotional datasets.

5.2.1 Diversity and generalizability

In order to improve generalization and promote AI fairness, it is crucial to ensure diversity while curating a multimodal dataset (Varsha 2023). Diversity should be considered across multiple dimensions including data sources, surroundings and environmental settings, modalities and affect categories, and participant demographics such as gender, age, ethnicity, dialects/accents, and cultural contexts. By emphasizing participant diversity, it can mitigate common problems such as gender bias, facial bias, age bias, annotation bias, skin and tone disparities, and racial discrimination (Xu et al. 2020; Domnich and Anbarjafari 2021; Li and Deng 2022b). Annotation bias is a significant concern as subjectivity can lead to inequalities and misinterpretations. By accurately reflecting the complexity of human experiences and emotions, we can create AI that is not only more equitable but also better suited to understanding diverse populations (Roshanaei 2024). The intended applications determine the significance of the specific diversity attributes needed in the dataset. In (Rathi and Tripathy 2024), the authors scrutinized diversity of various speech corpora and their application in emotion recognition. Applications pertaining to particular groups of citizens may not require ethnic diversity. Blanco-Ruiz et al. (2020) evaluated the impacts of gender on emotion elicitation under audiovisual stimuli. They reported that males and females have similar positive feelings and emotions while they have different negative emotions, especially for fear and contempt. Available datasets vary also in participants' ages. For example, their ages range between 19 and 27 years in (Kaur et al. 2018), from 21 to 33 years in (Livingstone and Russo 2018), and from 18 to 27 years in (Gnjatović and Rösner 2008). The recording settings and different devices used for data gathering are also included in the diversity of multimodal datasets. This entails taking into account elements that might greatly affect the quality and representativeness of the data, such as background noise, lighting, and the technology being used. User-generated datasets inherently embody this diversity, as each participant utilizes distinct devices, configurations, and recording environments (Morency et al. 2011; Pérez-Rosas et al. 2013b; Zadeh et al. 2016). However, datasets recorded in controlled environments often rely on standardized settings, with minimal attention given to the diversity of recording setups. This limitation is largely due to the associated costs and logistical challenges of altering environments and configurations (McKeown et al. 2010; Wu and Liang 2011; Katsigiannis and Ramzan 2018; Ab. Aziz et al. 2023). As a result, such datasets may lack the variability necessary to capture the full range of conditions that could enhance their applicability in real-world scenarios. Thus, it is important to address these issues in order to help development of more inclusive and robust emotional AI systems.

5.2.2 Naturalness and authenticity

According to Ekman and Friesen (1984), “an emotion can only be expressed genuinely if the stimulus is natural.” Natural emotional expressions are more spontaneous and exhibit

higher ecological validity to represent real-world emotions. However, collecting such naturalistic datasets in real-world environments remains a challenging endeavor (Dhall et al. 2012). In contrast, acting datasets are easier to collect, more controlled, and often contain more intense emotional states. The risk is that emotions may be exaggerated, biased towards skilled actors, or oversimplified (Lefter et al. 2017). Furthermore, naturalistic datasets can incorporate idiosyncratic emotion, where the same stimulus can trigger entirely different emotional responses by different people. K-EmoCon (Park et al. 2020) is the first publicly available multi-modal dataset that captures continuous emotions during naturalistic social interactions. As a result, the performance of emotion recognition significantly decreases when the automatic recognition models developed by such acted or induced datasets are used in real-life applications (Douglas-Cowie et al. 2003).

User-generated multimodal datasets are inherently organic, but come with notable limitations, particularly regarding diversity and clarity. They often require extensive manual preprocessing, which can be time-consuming and increase the likelihood of errors. For example, many YouTube videos contain an introductory sequence that shows a title, sometimes accompanied by visual animations. This issue was addressed by manually segmenting the video until the beginning of the first opinion utterance (Al-Azani and El-Alfy 2020). Additionally, video reviews may cover multiple topics, which was addressed by normalizing all video sequences to approximately 30 s in length (Morency et al. 2011). Furthermore, these datasets typically require people to face the camera without considering surrounding objects or events. Many user-generated videos feature specific objects and events that are not leveraged in models trained on such datasets. The researchers (Jiang et al. 2014) observed that certain emotions correlate strongly with specific semantics, such as the presence of particular events or objects. For example, the emotion of “joy” is often linked to events like parties or children playing. This insight inspired the researchers to investigate the use of semantic attributes as high-level representations for video emotion analysis. Their findings revealed that models based solely on attribute features achieved highly competitive performance. Furthermore, integrating attribute features with audio-visual elements enhanced the overall accuracies of individual models by up to 5%.

When assembling acted datasets, it is crucial to take into account the practice of conducting long-duration recordings. Sociolinguistic research underscores that, even in formal situations, subjects tend to ease into a more relaxed state after an hour, leading to freer and more expressive communication (Douglas-Cowie 1978). In addition, subjects should be skilled or expert actors (Sapiński et al. 2018; Perepelkina et al. 2018; Gunes and Piccardi 2006). For example, IEMOCAP dataset (Busso et al. 2008) relied on seven professional actors and three senior students from the Drama Department at the University of Southern California. The participants selected in (Chou et al. 2017) possessed real-life experiences in professional acting performances and were all students enrolled in the dramatic arts program at the National Taiwan University of Arts. Similarly, professional actors were recruited in (Sapiński et al. 2018) and semi-professional actors were involved in RAMAS (Perepelkina et al. 2018) for analyzing movements in emotional states. Moreover, it becomes occasionally necessary to organize structured interview sessions with participants. The interviewer or investigator can leverage prior knowledge of the subjects to customize specific conversations with the relevant participants (Douglas-Cowie et al. 2000). Posed vocal or facial expressions are often used in various contexts such as acting, social interaction or photography, where the person deliberately creates exaggerated specific emotions, e.g. smile, with more prototypical fea-

tures. Using an assembled dataset of 1877 voice clips, researchers conducted a comparative analysis of spontaneous and posed vocal expressions of emotions (Juslin et al. 2018). Their findings revealed that genuine emotional states are more readily conveyed and detected in spontaneous vocal expressions compared to posed expressions. Immersive virtual reality technology has been recently introduced allowing users to have a profound sense of presence and engagement into a three-dimensional computer-generated environments, which can intensify their emotional experiences. However, the authors of (Lima et al. 2024) found no substantial difference between the effectiveness of immersive and non-immersive virtual reality environment in detecting emotional responses to the presented stimuli. The above discussion emphasizes the need for balancing the tradeoff between naturalness and control while developing multimodal emotional datasets by combining diverse data collection strategies to fully capture the complexity and granularity of human emotional expressions.

5.2.3 Reliability and standardization

Reliability is a fundamental concept in statistics and psychometrics, and it specifically refers to the degree to which a test, experiment, or measurement instrument produces consistent, and stable results when repeated under similar conditions (Furr 2021), (Petersen 2024, Chapter 3). It can be categorized as test-retest reliability (stability over time), inter-rater reliability (consistency among annotators), internal consistency (consistency of related measures), and parallel forms reliability (agreement between different versions of the dataset). In the context of multimodal emotion recognition, reliability is crucial for ensuring accurate emotion detection and reproducibility/replicability of results when the collected data are analyzed multiple times, which can help informed decision-making processes. In multimodal emotion datasets, reliability is influenced by several critical factors, including naturalness of emotional expressions, consistency among raters/annotators, diversity of emotional states, contextual variability, modalities included, sensor accuracy, and stimuli employed to elicit emotions. Research indicates that the reliability of multimodal data is a key determinant of the model effectiveness and fusion algorithms (Zhang et al. 2024b). However, maintaining dataset reliability poses several challenges to mitigate the influence of noise, address imbalances in emotional state representation, handle incomplete data across modalities, and manage variation in data quality over time. Thus, the reliability of the multimodal emotional AI datasets also requires developing large-scale and high-quality datasets to ensure accuracy, consistency, reproducibility, and generalizability across diverse real-world scenarios for several reasons, including:

- Diversity of emotional states and expressions: Emotions are nuanced and diverse, and a larger dataset increases the likelihood of capturing a wide range of emotional states and sentiment expressions. Moreover, emotional expressions can vary significantly among individuals, cultures, and situations. A large dataset helps handle this variability, enabling AI models to adapt and generalize well to different contexts. Thus, the dataset should contain samples that represent all such factors. In particular, integrating physiological data can provide more objective continuous indicators that complement other modalities. Thus, they offer a valuable means of corroborating subjective ratings, and address the key limitation of self-reported and sentiment-based measures. For example,

an ultimate goal for VideoEmotion (Jiang et al. 2014) was to collect at least 100 videos for each emotional state.

- Model generalization: Large-scale datasets contribute to better generalization of models. With a diverse and extensive dataset, AI models can learn to recognize emotional cues across a wide spectrum of contexts and individuals, leading to more robust and accurate emotional understanding (Raudys et al. 1991). on the other hand, training sophisticated multimodal models, especially those combining text, image, and audio data, requires a substantial amount of diverse data to capture the complex interplay of modalities in expressing emotions effectively. There is a prevailing trend towards crafting multimodal sentiment and emotion analysis systems based on deep learning methodologies (Poria et al. 2017). It's crucial to note that deep learning techniques demand significantly larger datasets compared to traditional machine learning approaches.
- Real-world applicability: Emotional AI systems trained on large-scale datasets are more likely to perform well in real-world scenarios where emotional expressions can be unpredictable and varied.

Furthermore, achieving reliability in multimodal datasets necessitates systematic annotation by experts, taking the following aspects into account. For acted datasets, the emotional labels should not be assigned based on pre-defined emotional states of the emotional content to be acted, but human evaluators should be involved to assess the emotional content. In addition, the evaluation of the acted emotional content should not be performed only by the acting subject but also by skilled evaluators such as AMHUSE (Boccignone et al. 2017). Annotators should be professionals in the concerned language (Busso et al. 2008; Poria et al. 2018), have high level of education and from several knowledge backgrounds (Pereira et al. 2016). The annotators need to be briefed about the annotation process with a few examples before starting the annotation (Poria et al. 2018). It should be also multiple-layer annotations. For instance, the more familiar a face becomes, the more meaning you can discern from it. This might degrade the annotations' subjectivity and get more convergent results (Afzal and Robinson 2011). For example, the initial inter-annotation agreement for MUSTARD (Castro et al. 2019) was very low. Thus, the annotation process was stopped and the annotation differences were reconciled before proceeding further. The disagreements were discussed for a subset of 20 videos, and the videos were re-annotated. This notably improved the inter-annotator agreement as measured by the Kappa score from 0.1463 to 0.2326. In (Siegert et al. 2014), the authors investigated the inter-rater reliability for emotion annotation using Krippendorff's alpha. Other statistical measures of reliability include kappa statistics, intra-rater correlation coefficient, and concordance correlation coefficient.

5.2.4 Documentation and accessibility

This is indeed an essential requirement providing detailed description of the multimodal dataset, including format and organization of the dataset, collection methodology and data sources, participants' demographics, labeling/annotation process and quality, document changes between versions, usage guidelines and citation requirements, any data cleaning procedures, and any additional restrictions.² Detailed documentation or information provided about the multimodal dataset offer a comprehensive understanding of characteris-

²<https://dmeg.cessda.eu/>

tics and limitations to ensure long-term usage. An illustrative example of typical meta-data describing a dataset is shown in Table 7. Table 8 provides links for the publicly available multimodal datasets. In the following, we concentrate on three important aspects:

A) Participant demographics: Providing detailed demographic information about the participants is also important, including age, gender, cultural background, and any relevant characteristics. This also includes indicating how participants were selected and recruited. It is easier to provide those attributes for acted and induced datasets than user-generated datasets. For natural datasets, this can be performed by tracking the speakers metadata. For example, the demographics of the speakers in (Azani and El-Alfy 2019; Al-Azani and El-Alfy 2019a, 2020) were identified by looking for the participants' profiles. Therefore, selecting famous speakers is preferable for various reasons (Al-Azani and El-Alfy 2020). Firstly, retrieving their information from their profiles is more straightforward. Furthermore, the content of these renowned individuals holds more influence compared to other speakers. It could also pave the way for exploring new research directions, such as analyzing their perspectives over time. A recent study was conducted to analyze the impact of age and gender in sentiment analysis by (Kumar et al. 2020). It reported that the sentiment analysis model generated using female data was more accurate and the model generated for the senior users whose age above 50 years achieved better accuracy as compared to all other age groups. Therefore, providing participants' demographics is essential in many situations (Al-Azani and El-Alfy 2020).

Table 7 Typical attributes for multimodal dataset description

Total no. videos
Length of videos (hh:mm:ss)
No. of topics
Total no. videos expressed by male
Total no. videos expressed by female
Total no. distinct speakers
Total no. of samples per class
Average no. opinion segments in video
Average length of opinion segments/video (hh:mm:ss)
Average no. of video frames
Average word count per segment/video
Total no. words
Total no. unique words
Total no. words appears at least n times in the dataset
of dialogues
Avg. # of utterances per dialogue
Avg. # of emotions per dialogue
Avg./Max # of speakers per dialogue
of emotion shift
of stimuli
Type of Stimuli used: Audio, visual, tactile, olfaction, haptic, driving task, cognitive assessment task
Number and types of used devices
For physiological signals
ID; AGE; SEX; PROCEDURE START TIMESTAMP; PROCEDURE END TIMESTAMP; BIOSIGS START TIMESTAMP; BIOSIGS END TIMESTAMP; LABEL; PROCEDURE; COMMENTS

Table 8 Publicly available multimodal datasets and their primary tasks

Dataset	Task	Link
eNTERFACE (Martin et al. 2006)	Emotion	http://www.interface.net/interface05
IEMOCAP(Busso et al. 2008)	Emotion	https://paperswithcode.com/dataset/iemocap
VAM(Grimm et al. 2008)	Emotion	http://emotion-research.net
ICT-MMMO (Wöllmer et al. 2013)	Sentiment	By sending mail to Giota Stratou (stratou@ict.usc.edu)
MOUD (Pérez-Rosas et al. 2013b)	Sentiment	https://web.eecs.umich.edu/~mihalcea/downloads.html
RECOLA(Ringeval et al. 2013)	Emotion	https://diuf.unifr.ch/main/diva/recola/download.html
News Rover (Ellis et al. 2014)	Sentiment	http://www.ee.columbia.edu/lndvmm/news-rover/sentimentdataset/
VideoEmotion(Jiang et al. 2014)	Emotion	https://drive.google.com/drive/folders/0B5peJ1MHnIWGd3pFbzMyTG5BSGs
SWELL-KW(Koldijk et al. 2014)	Emotion, stress	https://cs.ru.nl/~skoldijk/SWELL-KW/Dataset.html
MOSI (Zadeh et al. 2016)	Sentiment	http://immortal.multicomp.cs.cmu.edu/raw_datasets/
MSP-IMPROV(Busso et al. 2016)	Emotion	https://ecs.utdallas.edu/research/researchlabs/msp-lab/MSP-Improvement.html
EMOREACT(Nojavanasghari et al. 2016)	Emotion	https://github.com/bnojavan/EmoReact
MOSEI(Zadeh et al. 2018)	Sentiment + Emotion	http://immortal.multicomp.cs.cmu.edu/raw_datasets/
MELD(Poria et al. 2018)	Sentiment + Emotion	https://affective-meld.github.io/
WESAD(Schmidt et al. 2018)	Stress, amusement	https://www.eti.uni-siegen.de/ubicomp/home/datasets/icmi18/index.html.en?lang=en
MMSGDS(Al-Azani and El-Alfy 2019b)	Sentiment, gender	By contacting the corresponding author
Piim News(Pereira et al. 2019)	Emotion	http://www.icwsm.org/2016/datasets/datasets/
MUStARD(Castro et al. 2019)	Sarcasm	https://github.com/soujanyaoria/MUStARD
BIRAFFE(Kutt et al. 2019)	Personality	https://zenodo.org/record/3922055
ElderReact(Ma et al. 2019)	Emotion	https://github.com/Mayer123/ElderReact
SADAM(Al-Azani and El-Alfy 2020)	Sentiment, Demographics	By contacting the corresponding author
K-EmoCon(Park et al. 2020)	Emotion	https://zenodo.org/record/3931963
ECSMP(Gao et al. 2021)	Emotion,cognition, sleep quality	https://data.mendeley.com/datasets/vn5nknh3mn/2
BIRAFFE2(Kutt et al. 2022)	Personality	https://zenodo.org/record/5786104
MGEED(Wang et al. 2023)	Emotion	https://github.com/YMPort/MGEED
StressID (Chaptoukaev et al. 2023)	Emotion, Stress	https://project.inria.fr/stressid/

B) Data collection procedure and recording environment: A comprehensive description of the procedure, collection, and recording environment is foundational for multimodal emotional AI datasets to ensure reproducibility, methodological transparency, bias-free and fairness, and ethical standards. The physical setting where data was recorded, including lighting conditions, background, and any controlled variables, needs to be described. In

addition, the type of equipment used for data capture also should be specified including cameras, microphones, and physiological sensors. For data collected from social media or the web, the search process and the inclusion/exclusion criteria or any constraints should be described (Morency et al. 2011).

C) Description and organization: Another issue that obstructs the deployment of the multimodal datasets is the lack of the perfect organization and description. The data files need to be standardized and transparent (Douglas-Cowie et al. 2003). Different formats may be utilized such as video files (e.g. 'mp4', 'mpeg-4'), audio files (e.g. 'wav', or 'mp3'), and text and xml files (e.g. '.txt' or '.xml'). Additionally, it can have physiological signals (e.g. 'csv' or 'mat'), and image files (e.g. 'png', or 'jpeg'), if applicable. This is in addition to other meta-descriptions of the whole dataset, which may include sample_ID, segment_ID, the time stamps for each segment of the videos/audios, the corresponding script (raw and cleaned text), the speaker/participant demographics, etc such as (Alowisheq et al. 2023). As an example of physiological datasets, BIRAFFE (Kutt et al. 2019) dataset attributes include id, age, sex, procedure-begin-timestamp, procedure-end-timestamp, biosigsbegin-timestamp, biosigs-end-timestamp, openness, conscientiousness, extraversion, agreeableness, neuroticism, neo-ffi, biosigs, etc.

5.2.5 Ethical considerations and compliance

Collecting and experimenting with datasets of emotional material are surrounded by several ethical, copyright, and privacy issues as they involve sensitive personal information (e.g., facial expressions, physiological signals, voice data) (Douglas-Cowie et al. 2003; Serban et al. 2018; Mohammad 2022). Katirai (2023) defined three major ethical issues related to emotion recognition technologies: (i) the risk of biased and inequitable outcomes stemming from the flawed foundations and questionable assumptions on which these technologies are built, (ii) the privacy concerns surrounding the sensitive emotional data processed by these systems, and (iii) the potential harm that could arise from the deployment of these technologies in sensitive domains such as employment, education, healthcare, and law enforcement. Ethical considerations such as licensing restrictions and consent agreements have significantly impacted distribution of multimodal emotional datasets (Hu and Flaxman 2018), prohibiting open access, limiting accessibility for research purposes, hindering cross-institution collaboration, posing challenges on reproducibility, reducing opportunities for benchmarking, and subsequently slowing progress in emotional AI research.

Obtaining informed written consent from all participants is essential before any data collection, particularly when using multiple modalities to assess emotional states. This consent should clearly explain the purpose of the data collection, the use of AI for emotion recognition, the procedure involved, any potential healthcare applications, and any restrictions on data redistribution. Clear communication ensures transparency and supports ethical practices in the use of personal data (Katsigiannis and Ramzan 2018; Livingstone and Russo 2018; Miranda et al. 2022; Kutt et al. 2022, 2019). All universities of standing have Research Ethics Boards to ensure experiments involving the collection of human data are conducted in accordance with national and international law (Weinbaum et al. 2019; Al-Azani et al. 2022). Some of the reviewed studies obtained approvals from specific committees in universities before conducting the studies or collecting the data (Park et al. 2020; Katsigiannis and Ramzan 2018; Kutt et al. 2022; Gao et al. 2021). For example, the protocol for collecting

K-EmoCon (Park et al. 2020) dataset was reviewed and approved by the Korea Advanced Institute of Science and Technology Institutional Review along with the written consent of the participants. Additionally, the consent form involved five issues: "(1) the objective of gathering data, (2) the procedure for collecting data, (3) types of data to be collected from participants, (4) compensation plan for participation, and (5) the protocol for the protection of privacy-sensitive data". Other issues to be explained may include the impact of speakers' appearance on the public while expressing opinions, feelings, or appraisals. Other restrictions may be included such as the collection of DSDRWDT (Healey and Picard 2005) required each driver to have a valid driving license besides signing a formal consent form. For medical research involving human subjects, the declaration of Helsinki (Shrestha and Dunn 2019) provides an example of ethical guides that were followed in some studies such as (Livingstone and Russo 2018).

Mohammad (2022) listed a set of examples of inappropriate commercial automated emotion recognition applications related to misused and abused emotional AI systems. Several techniques and approaches have been proposed and explored to ensure security and privacy while training multimodal AI models, including differential privacy, federated learning, homomorphic encryption, and swarm learning (Acosta et al. 2022). A recent survey by Shahriar et al. (2023) defined several risk issues and solutions related to individual's privacy and their impacts on ethical processing and protection of data during the AI life cycle (planning, data collection, data preparation, model development, and model deployment). These issues include:

- i. Risk of identification: refers to any action that may threaten the disclosure of the human subject's identity.
- ii. Risk of making an inaccurate decision: accuracy is crucial for a trustworthy AI system as inaccuracies can lead to harmful social, political, and legal outcomes.
- iii. Risk of non-transparency in AI systems: refers to the need for automated data profiling and decision-making to be transparent, explainable, and free from bias and discrimination.
- iv. Risk of non-compliance with privacy regulations: refers to the fact that inadequate compliance with privacy regulations and policy recommendations poses a risk to privacy and can have significant consequences for organizations.

We recommend utilizing a card, as depicted in Fig. 8, to prompt the subject to provide their information and also as an agreement for distributing the dataset to the research community. The proposed card is composed of two main parts the top part which covers the subject/participant details, including her/his ID in the dataset, and the name. This part also includes the demographic information of the subject, including gender, Birth date, Education level Occupation, Income average (per month or per year), and Marital status. Other demographic attributes might be added based on the objective of the dataset. The bottom part, "Data Usage Agreement", includes the agreement statement for the purpose of data collection and distribution. Other options might be added.

Fig. 8 A sample of an actor agreement card for data collection, storage, use and redistribution

Subject ID:		
First Name: Middle: Last:		
Gender: ☐ Male ☐ Female		Birthdate: DD/MM/YYYY
Education Level: ☐ early childhood ☐ primary ☐ secondary ☐ bachelor ☐ master ☐ doctor		Nationality:
Occupation:		Income average (monthly):
Marital status : ☐ Single ☐ Married ☐ Divorced		
Email:		Mobile:
I amand I freely give my consents for the collection, storage, and use of my emotional related data recorded on DD/MM/YYYY, exclusively for RESEARCH purpose, and I understand the implications of this consent statement, with the following CONDITION/S: ☐ I agree that my collected data can be disseminated in publications. ☐ Non-anonymized. ☐ Anonymized. ☐ I agree that my collected data can be disseminated to a third party for reuse in research purpose. ☐ Non-anonymized. ☐ Anonymized. Date: DD/MM/YYYY Signature:		

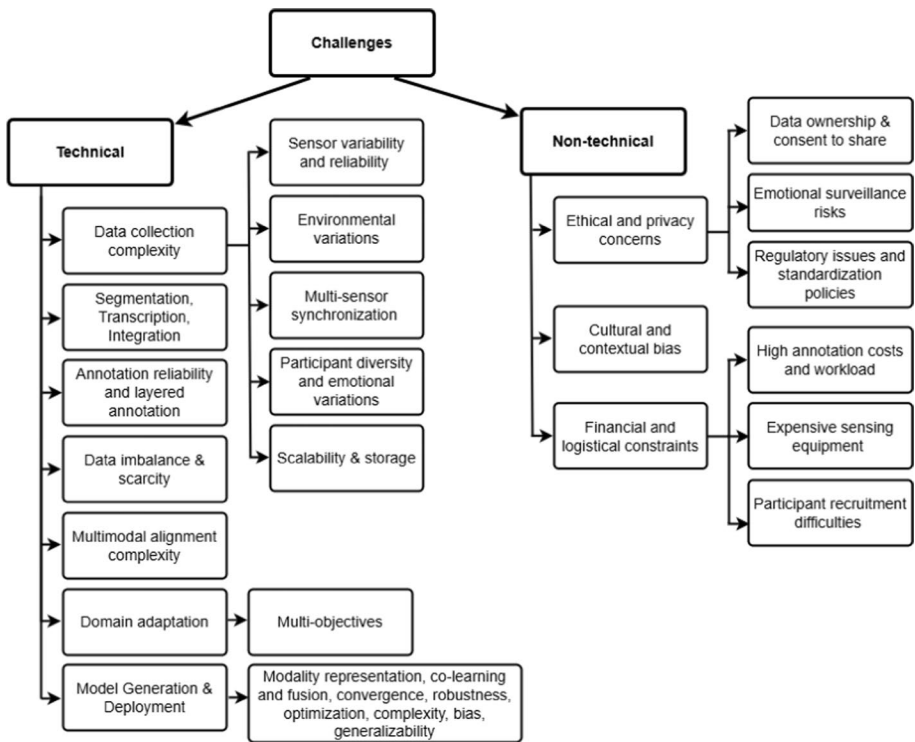


Fig. 9 Challenges of multimodal datasets

5.3 Challenges

Several challenges are highlighted in this section and are categorized into two main groups: non-technical and technical challenges, as illustrated in Fig. 9. Non-technical challenges pertain to regulatory and procedural issues that naturally arise during the dataset construction process. They involve aspects related to human and society such as culture and con-

textual variations, subject selection and recruitment, group diversity and skewness, privacy concerns, regulatory and standardization aspects. Other issues include data collection costs and workload, type of environment (e.g. natural spontaneous, acted, induced), and other financial and logistic aspects. On the other, the technical challenges are more about computational frameworks, systems, methods and algorithms throughout the pipeline of data collection, cleaning/cleansing, and processing. They involve aspects such as sensor variability and reliability, multi-sensor synchronization, storage and scalability, multimodal alignment and fusion, data imbalance and scarcity, augmentation, domain adaptation, as well as model generation, co-learning and deployment challenges. In the following, we discuss several of these aspects in more details.

5.3.1 Subject selection challenges

The process of selecting subjects or participants for multimodal datasets is not a straightforward task. It involves several considerations to ensure a representative and diverse sample related to the research objectives, subjects' demographic characteristics or traits, and meeting inclusion and exclusion criteria of the study. For example, the participants in the acted NNIME dataset (Chou et al. 2017) were selected to have prior real-life experiences in professional acting, hence they were students in the dramatic arts degree program at the Department of Drama at the National Taiwan University of Arts at the time of recording. There were 22 female and 20 male students with ages range from 19 to 30. Moreover, they were native Mandarin Chinese speakers.

5.3.2 Data ingestion challenges

The first step in the dataset development where data is collected, recorded, and transferred to a storage or processing system. The procedure and environment settings for multimodal emotional AI datasets involve systematic and controlled methodologies for gathering data across various modalities, including text, images, audio, and physiological signals. This encompasses the establishment of protocols and conditions to capture a diverse range of emotional expressions while maintaining consistency, reliability, and ethical standards throughout the data collection process. The environment settings address physical and contextual conditions such as lighting, sound, and participant interactions, ensuring the quality and relevance of the collected multimodal data, including physiological signals, for comprehensive emotional analysis. Table 9 outlines various recording procedures and scenarios for the multimodal datasets.

5.3.3 Preprocessing challenges

Preprocessing is a key process in the entire workflow of data engineering for preparing high-quality, reliable, noise-free, standardized and consistent data for analysis and training AI models. It involves data integration, data cleaning and cleansing, segmentation, and transformation. Key preprocessing steps and methods vary depending on the nature of the modality under processing, e.g. 2D/3D image, video, or time series. They can be applied on individual modalities or cross modalities. Table 10 outlines examples of relevant modality-specific preprocessing tasks for different modalities and potential challenges faced.

Table 9 Protocols available to collect and record multimodal emotional AI datasets

Protocol	Task	Modality	Ref
A single individual is speaking directly in front of the camera, expressing their opinion on a specific topic of interest	Sentiment, Emotion, Demographics	Video	(Morency et al. 2011), MOSI (Zadeh et al. 2016), AMMD (Alqarafi et al. 2017), SADAM (Al-Azani and El-Alfy 2020)
Asking subjects to read, memorize and pronounce emotional experiences	Emotion	Video	(Caridakis et al. 2010; Bänziger and Scherer 2010), eNTERFACE (Martin et al. 2006), (Pápay et al. 2011)
Asking subjects to improvise based on hypothetical scenarios	Emotion, Sentiment	Audio, textual	(Wu and Liang 2011)
Wizard-of-Oz (WOZ) scenario; interactions between users and an operator (HCI)	Emotion	Audio, visual	(Gnjatović and Rösner 2008; McKeown et al. 2010)
Watching or listening segments of movies, pictures, or music	Emotion	Audio, visual, textual, physiological	(Soleymani et al. 2012; Koelstra et al. 2012), DECAF (Abadi et al. 2015), DREAMER (Katsigiannis and Ramzan 2018), MGEED (Wang et al. 2023), AMI-GOS (Miranda-Correa et al. 2018), ASCERTAIN (Subramanian et al. 2016), SenseEmotion (Velana et al. 2017), ECSMP (Gao et al. 2021)
Recording (in videos) subjects' engagements when watching Massive Open Online Courses (MOOCs) video material and then asking them to express their views about the videos	Emotion	Visual	MOOCs (Kaur et al. 2018)
Dyadic/multiparty conversational dialogue	Emotion	Visual, audio, signals	NNIME (Chou et al. 2017), RECOLA (Ringeval et al. 2013), K-EmoCon (Park et al. 2020)
Facial Action Coding System (FACS): The subjects were instructed to display N series of single or combined expressions defined in FACS	Emotion	Visual	(Ekman and Friesen 1978; Ekman 2002)
Driving	Stress	Visual, physiological	DSDRWDT (Healey and Picard 2005)
Cognitive testing, education and research tasks	Emotion, stress	Physiological signals, video, audio	ECSMP (Gao et al. 2021), (Kaur et al. 2018), SWELL-KW (Koldijk et al. 2014), WESAD (Schmidt et al. 2018), (Nojavanasghari et al. 2016), ElderReact (Ma et al. 2019), StressID (Chaptoukaev et al. 2023)
Heat stimulation	Emotion, pain	physiological signals	SenseEmotion (Velana et al. 2017)
Game sessions: participants playing cooperative (co-op) and competitive board games	Emotion	Visual	MP-BGAAD (Schimmel et al. 2019), MUMBAI (Doyran et al. 2021), (Wu and Liang 2011)

Table 9 (continued)

Protocol	Task	Modality	Ref
The participants receive instructions on the experiment without specific details on the purpose of the study, to avoid biased reactions	Emotion, amusement	video, physiological responses	AMHUSE (Boccignone et al. 2017)

Data integration combines recorded and streamed data from multiple sources. For example, certain retrieved or collected videos may not align with the dataset objectives or meet the inclusion criteria, necessitating their removal. These criteria vary depending on the dataset type and research objectives. For instance, in studies such as (Morency et al. 2011; Zadeh et al. 2016; Alqarafi et al. 2017; Al-Azani and El-Alfy 2020), specific conditions apply, e.g. the speaker should be positioned in front of the camera, with their face clearly visible and minimal occlusion. Furthermore, recordings should be devoid of background music or animation.

Data cleaning and cleansing is the process of identifying and correcting inaccuracies and inconsistencies in a dataset to improve its quality and reliability. Although sometimes these two terms are used interchangeably, they refer to different scopes of data quality process. While data cleaning works at a micro-level to correct individual record errors and formatting issues, remove duplicates, as well as handle missing values, data cleansing works at a broader macro-level approach to ensure data reliability, consistency, enrichment, and standardization across entire datasets and sources (Hosseinzadeh et al. 2023).

Segmentation is another preprocessing step that divides continuous multimodal data streams into meaningful emotional segments or chunks. For example, partitioning videos into sequenced frames and spoken segments, ensuring that each segment expresses a state, mood, sentiment, or emotion. The splitting process can be time based, e.g. segmenting data into fixed or variable-length intervals. Segments can represent utterances based on long pauses in (Morency et al. 2011). Each utterance does not need to represent a separate sentence. For example, in (Castro et al. 2019) 38.7% of utterances consist of two or more sentences while 61.3% of the utterances are single sentences. Splitting can also be based on detected emotional transitions or changes (e.g. transition from neutral to angry or happy) or based on modality characteristics (e.g. splitting speech transcript at sentence boundaries or splitting EEG signals based on brainwave shifts). Zadeh et al. (2016) have established specific rules to ensure precise segmentation, such as 1) segmenting subjective content based on the number of disclosed private states, 2) segmenting if the utterance involves a modification of a private state while retaining the subject, and 3) segmenting if the subjective utterance concludes with the commencement of an objective segment. In the work by Douglas-Cowie et al. (2007), segments are chosen from the entire recording based on the criterion that the segmented clips should align with ratings derived from the maximum available recording quality. Different tools have been used for segmentation. For example, PySceneDetect tool (Castellano 2023) was used to automatically segment ElderReact dataset (Ma et al. 2019) videos into short clips. Other tools such as Praat and OpenEAR (Eyben et al. 2009), and ELAN (Brugman et al. 2004) have been utilized for segmentation. Each tool is optimized for certain modalities. While PySceneDetect excels in automated scene detection, it is less effective for emotion driven segmentation. Praat is superior for speech

Table 10 Per-modality preprocessing tasks and challenges

Modality	Preprocessing Task	Methods	Challenges	References
Audio	Data cleaning/cleansing	Noise reduction (Wiener filtering, spectral subtraction, Wave-U-Net, SEGAN)	Background noise variation, loss of low-energy speech	(Davis 2018; Cherukuru and Mustafa 2024)
		Outlier removal, silence trim, distortion filter	Speech variability, overlapping speakers	
	Segmentation	Voice activity detection, Speech pause detection, speaker diarization	Overlapping speech, Misclassification of silence as speech, misclassification of speakers	
	Time alignment	Dynamic Time Warping (DTW) with Phoneme Matching	Emotion transition mismatch in aligned speech segments	
	Feature extraction/learning	MFCCs, spectrograms, prosodic features, CNNs, Transformers	Loss of subtle emotional cues in frequency compression	
	Normalization	Z-score, Min-max scaling	Speaker-dependent pitch variations	
Image	Data cleaning/cleansing	Denoising filters (Gaussian, Median), deblur, GAN-based noise removal	Risk of removing fine-grained emotional expressions, occlusions and background clutter	(Archana and Jeevaraj 2024; Woods and Gonzalez 2019)
		Duplicate image detection, background removal	Variability in face angles and expressions, occlusion	
	Image enhancement	Super-resolution networks, denoising autoencoders	Heavy computational load caused by complex networks	(Hsu et al. 2024; Umirzakova et al. 2024; Gao et al. 2025; Rizvi et al. 2025)
	Segmentation	Scene segmentation, face cropping, face landmarks, skeleton, MTCNN, RetinaFace, ViT	Incorrect face alignment in low-resolution images, occlusion, lighting variations	
	Feature extraction/learning	HOG, SIFT, CNNs (ResNet, EfficientNet), ViT	feature loss due to pose and lighting variations, occlusion	
	Normalization	Gamma correction, contrast and brightness normalization, adaptive histogram equalization	Over-enhancement leading to unnatural image artifacts, noise and outliers impact	

Table 10 (continued)

Modality	Preprocessing Task	Methods	Challenges	References
Video	Data cleaning/cleansing	frame duplication removal, motion blur, motion smoothing, filtering, frame rate adjustment	Motion artifacts, frame inconsistencies, occlusion, lighting variations, removal of fine-grained emotion cues, background changes	(Khanna et al. 2024; Tekalp 2015)
		Detect shot boundary, extract keyframes	Difficulties in emotion transition detection, background changes, removal of fine-grained emotion cues	
	Segmentation	Detect scene change, optical flow, Pose estimation, RNNs, CNNs, ViT	Complex background motion interfering with segmentation	
	Feature extraction/learning	Optical flow, CNNs, RNNs, ViT, OpenPose	Feature loss due to low resolution, background changes, ambiguity, artifacts	
	Transformation	Frame interpolation, GAN-based upscaling, optical flow, attention based	Loss of original emotion details, distinguish fine-grained emotion changes	
Text	Data cleaning/cleansing	Spell checking, unicode normalization, encoding fixes, removing non-language chars, BART	Handling informal language, typos, dialects, loss of original intent	(Lauriola et al. 2022; Fanni et al. 2023)
		Duplicate sentence removal, ambiguity resolution	Loss of meaningful words or phrases, loss of original intent	
	Segmentation	sentence split, phrase chunks, tokenization	Misalignment in long and complex sentence structures	
	Normalization	Lemmatization, stemming, lowercasing	Over-normalization may alter text semantics	
	Feature extraction/learning	Word embeddings (Word2Vec, FastText, BERT, RoBERTa), emotion lexicons	Context and intent loss in vector-based text representation	

Table 10 (continued)

Modality	Preprocessing Task	Methods	Challenges	References
Physiological	Data cleaning/cleansing	Denosing/filtering, artifact removal (ICA, PCA)	Physiological noise from movement, blinking	(Cruz-Vazquez et al. 2025 ; Bota et al. 2019 ; Robbins et al. 2020)
		Missing data imputation, outlier removal	Variability in physiological responses	
	Alignment and synchronization	Dynamic time wrapping (DTW), transformers	need synchronization buffers, noise, missing values	
	Segmentation	Epoch-Based segmentation, event-related potentials	Defining meaningful time segments	
	Feature extraction/learning	Frequency band (Alpha, Beta, Gamma), ICA, PCA, CNNs, RNNs, attention-based	Feature drift due to sensor variations, implicit emotions	
	Normalization	Z-score, PSD normalization	Physiological data differs across subjects	
	Transformation	Wavelet, STFT	Computational complexity of transformation	

segmentation but not suitable for segmenting video scenes, and ELAN is best suited for multimodal data but requires manual annotation.

Data transformation aims to convert raw data into more structured, optimized and machine readable format. It may involve methods for encoding categorical variables, feature extraction and selection, discretization, scaling, standardization and normalization, and dimensionality reduction. Several examples are illustrated in Table 10 for each modality. Different data types (audio, image, video, text, and physiological) can lead to more challenges, e.g. modalities may have different numerical ranges, time rates, sensor errors, etc. Additionally, while developing multimodal datasets, some modalities may dominate than others like facial expressions versus physiological signals (Li et al. [2023](#)). Missing values or gaps within one or more modalities can also occur for various reasons such as sensor error or equipment malfunction, temporal obstructions, poor lighting, noisy environments, data corruption during transmission, or timing mismatches causing portions of modalities to be omitted. Handling missing, incomplete or imbalanced data within one or more modalities to ensure dataset completeness and robustness may require imputation or sophisticated preprocessing methods including deep learning techniques and generative AI (Lan et al. [2024](#); Baltrušaitis et al. [2018b](#); Ma et al. [2021](#); Wu et al. [2024](#)).

Besides handling each modality separately, preprocessing can also take advantage of transferring knowledge from other available modalities, leading to cross-modality preprocessing techniques (e.g. audio-to-text, image-to-audio, image-to-text). Examples of these

processing are cross-modality denoising, alignment, and synchronization. Misaligned modalities can adversely impact the model performance and interpretability. The alignment and synchronization process of heterogeneous data streams for emotion recognition involves coordinating and matching the temporal aspects of data from different modalities, such as audio, video, and textual information. This ensures that corresponding elements from each modality are properly timed and can be jointly analyzed or used for training machine learning models. The primary goal is to establish a coherent and temporally aligned dataset, enabling meaningful cross-modal relationships and interactions to be captured accurately. However, the alignment process can be a daunting task due to variations in sampling rates and time scales among different modalities. Various tools and techniques, such as traditional Dynamic Time Warping (DTW) and deep learning alignment methods, have been also employed to address this task (Xu et al. 2019a; Li and Tang 2024).

5.3.4 Transcription challenges

During this step, the spoken words are extracted and other meta information such as the start time of each spoken utterance. Transcription can be conducted either manually, automatically, or through a combination of both manual and automated methods. For some languages, there are some tools to perform it automatically such as IBM WatsonTM Speech-to-Text service, Automatic Speech Recognition (ASR), and Kaldi (Povey et al. 2011) tools. IBM WatsonTM Speech-to-Text service was applied in (Pereira et al. 2019). Some platforms such as Big Bang Theory provide subtitles or transcripts readily which was utilized in MUSTARD dataset (Castro et al. 2019). Transcription in a manual manner might be a tedious task and time-consuming, yet it is reliable. The textual features extracted from the manual and automated transcriptions generated using ASR were evaluated in (Wöllmer et al. 2013). It was reported that the performance was significantly and consistently decreased of roughly 10% absolute when applying texts generated using ASR. Speech transcription was performed manually in (Morency et al. 2011; Al-Azani and El-Alfy 2020). Transcription was conducted semi-automatically where a technique was applied to generate the text then the generated text was revised by humans. For example, transcription in (Wöllmer et al. 2013) was performed manually and using ASR. Other studies automatically generated transcriptions (Wöllmer et al. 2013; Rozgić et al. 2012; Govindaraj and Gopalakrishnan 2016). The transcription process should also be clearly outlined, encompassing the definition of transcript criteria. The transcripts for some datasets were prepared to represent pronunciation: murmurs, pauses, coughs, laughter, etc. by a phonetic alphabet, and the speech recordings were aligned with their textual interpretation (Cristea et al. 2020). Furthermore, the transcription scheme in (Zadeh et al. 2016) contains details about pause fillers (such as “umm” and “uhh”), stresses, and speech pauses. Utterances such as ‘yeah,’ ‘okay,’ and ‘no’ can convey diverse emotions based on the context and discourse of the dialogue (Poria et al. 2018). In (Busso et al. 2008) detailed motion capture information for the head, face, and to some extent, the hands, the start time of each spoken utterance was provided. In MUSTARD (Castro et al. 2019), each utterance is accompanied by a context of historical utterances within the dialogue to offer additional context about the scenario in which the utterance takes place.

5.3.5 Annotation and inter-rater agreement challenges

In the context of a specific task or application, where a set of labels is predefined based on the objectives, and a corresponding set of samples is available, the process of annotation becomes pivotal (Pustejovsky and Stubbs 2012). Annotation involves the meticulous task of assigning an accurate label to each sample within the dataset. In this process, every individual sample is carefully annotated or labeled, establishing a direct association with its respective class. This annotation procedure is foundational in the preparation of a labeled dataset, a critical resource essential for training and evaluating machine learning models (Devillers et al. 2005). The precision of this annotation process is paramount, as it lays the groundwork for developing robust and accurate models tailored to the specific task or application at hand. Annotation task for video datasets can be conducted at several levels: frame level, utterance, or video level.

Supervised machine learning approaches place a substantial reliance on extensive and well-labeled training datasets. However, generating such datasets is a resource-intensive and time-consuming process and is often constrained by scarcity. To address these challenges, online crowd-sourcing systems have been proposed as a viable solution. These systems facilitate the collection of annotations from diverse annotators, offering a more efficient and cost-effective alternative. By distributing complex tasks among numerous annotators, these systems have been utilized to expedite the creation of large-scale annotated datasets with reduced time and cost implications, as highlighted in the work by Zhan et al. (2019). Figure 10 provides a taxonomy for the annotation process of multimodal datasets and Table 11 summarizes the considered methods, tools, levels, and evaluators' requirements for multimodal datasets' annotation process.

It is important to select several professional annotators, and the annotators should be briefed about the annotation process with a few examples before starting the annotation in order to reduce the inter-rater inconsistencies (Poria et al. 2018). For example, in (Pereira et al. 2016), four professionals from several fields of knowledge (Journalism and Languages, Mathematical Modeling, Law, and Pedagogy) made manual annotations of the tension levels of the news videos. The number of annotators in NNIME (Chou et al. 2017) is 49 annotators contributing to a thorough and varied labeling process that improves the reliability and validity of the annotations. This is a high number compared to other reviewed datasets. For example, only two annotators have been involved in MOUD (Pérez-Rosas et al. 2013b), SMOD (Pérez Rosas et al. 2013a), and AFEW (Dhall et al. 2012). Other datasets such as KVDERW/KVDERW-Eng (Khanh et al. 2021) and RECOLA (Ringeval et al. 2013) were annotated by six individuals. In NNIME dataset, different types of annotators were involved: 1) 44 peer-reports: annotators with drama background 2) one director-report: professional director, 3) one self-report: actors of the interaction, 4) four observer-reports: naive observers with no explicit training in drama. To mitigate gender bias in attention, the annotation task in (Park et al. 2014) was conducted with a focus on gender. This approach involved evaluators exclusively assessing speakers of the same gender.

In addition, other scenarios have been utilized to assist the annotation tasks. Search terms were utilized for video annotation, a process referred to as weak labeling (Zhou 2018). For example, a list of positive and negative sentiment keywords was used in (You et al. 2016) to query Getty Images. The returned images were labeled based on the sentiment labels of the keywords. However, relying solely on the search terms from collected videos for

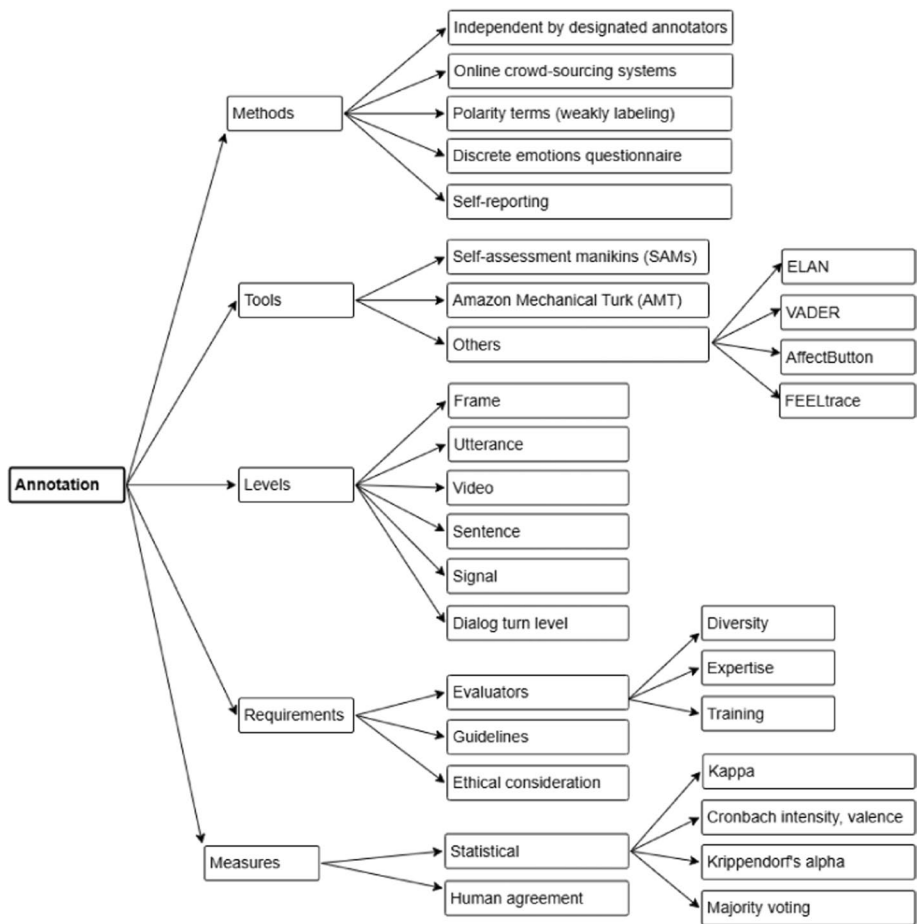


Fig. 10 A taxonomy for multimodal datasets' annotation

annotation tasks is not recommended. This is due to the potential inconsistency between the information annotated by users when uploading videos and the genuine emotional state, as highlighted in (Chen et al. 2021).

In the case of CHEAVD (Li et al. 2017), which introduced multi-emotion labels and fake/suppressed emotion labels, annotators were specifically instructed to annotate segments perceived as ambiguous based on facial expressions and voices. This approach aimed to capture nuanced emotions and instances of intentionally concealed or misrepresented emotions in the dataset.

The Discrete Emotions Questionnaire (Degottex et al. 2014) was employed as a framework for formulating questions posed to annotators during the annotation process for Elder-React (Ma et al. 2019). For instance, annotators were prompted with the question, “How angry/mad/frustrated was the individual in the video?” to assess the emotion of “Anger”. Expected responses for emotional intensity ranged from “Not at all” to “Very much”, corresponding to a 1-4 scale, respectively.

Table 11 Overview of the annotation framework for multimodal datasets

Category	Subcategory	Examples
Method	Independent annotation by designated annotators or professionals	Morency et al. (2011); Wöllmer et al. (2013); Pérez-Rosas et al. (2013b); Pérez Rosas et al. (2013a); Martin et al. (2006); Pereira et al. (2016, 2019); Zadeh et al. (2016); Wu and Liang (2011); Abburi et al. (2016); Busso et al. (2008); Jiang et al. (2014); Najadat and Abushaqra (2018); Alqarafi et al. (2017); Poria et al. (2018); Borth et al. (2013); You et al. (2015); Al-Azani and El-Alfy (2020)
	Polarity/emotional search terms (Weakly labeling)	You et al. (2016)
	Discrete Emotions Questionnaire	ElderReact (Ma et al. 2019), AMIGOS (Miranda-Correa et al. 2018)
	Self-reporting	Chou et al. (2017); Boccignone et al. (2017); Katsigiannis and Ramzan (2018); Ab. Aziz et al. (2023); Wu and Liang (2011), MGEED (Wang et al. 2023)
Tools	Video reviews rating	Pérez Rosas et al. (2013a), ICT-MMMO (Wöllmer et al. 2013)
	Self-assessment manikins (SAMs)	Busso et al. (2008); Ringeval et al. (2013); Wang et al. (2023); Miranda et al. (2022); Velana et al. (2017); Ringeval et al. (2013)
	Amazon Mechanical Turk (AMT)	Borth et al. (2013); You et al. (2016); Pereira et al. (2019); Poria et al. (2018); Zadeh et al. (2018); Nojavanasghari et al. (2016), ICT-MMMO (Wöllmer et al. 2013)
	ELAN	Pérez-Rosas et al. (2013b); Nojavanasghari et al. (2016); Schimmel et al. (2019); Doyran et al. (2021)
	VADER	You et al. (2016)
	AffectButton	Boccignone et al. (2017)
	FEELtrace	McKeown et al. (2010, 2011)
Levels	Frame	ICT-MMMO (Wöllmer et al. 2013)
	Utterance	eINTERFACE (Martin et al. 2006), YouTube (Morency et al. 2011), - (Wu and Liang 2011), MOUD (Pérez-Rosas et al. 2013), MOSI (Zadeh et al. 2016), MSP-IMPROV (Busso et al. 2016), AMMD (Alqarafi et al. 2017), MOSEI (Zadeh et al. 2018), MELD (Poria et al. 2018), MMSGDS (Al-Azani and El-Alfy 2019b), MUsTARD (Castro et al. 2019), M3ED (Zhao et al. 2022b)
	Video	ICT-MMMO (Wöllmer et al. 2013), SMOD (Pérez Rosas et al. 2013a), (Pérez Rosas et al. 2013a), Telugu (Abburi et al. 2016), StressID (Chaptoukaev et al. 2023)
	Sentence	News Rover (Ellis et al. 2014)
	Signal	DECAF (Abadi et al. 2015), DREAMER (Katsigiannis and Ramzan 2018), WEMAC (Miranda et al. 2022), StressID (Chaptoukaev et al. 2023)
	Dialog turn level	IEMOCAP (Busso et al. 2008), MUsTARD (Castro et al. 2019)
	Expertise	IEMOCAP (Busso et al. 2008) University of Southern California students who were fluent English speakers In (Pereira et al. 2016), four professionals from several fields of knowledge
Evaluators	Diversity	NNIME (Chou et al. 2017) 49 annotators K-EmoCon (Park et al. 2020): self-annotation, partner-annotation, and external observer annotation

For acted or induced datasets, self-reporting as the task of asking all actors to evaluate their performance at the end of each stimulus was utilized in different datasets such as in (Chou et al. 2017; Boccignone et al. 2017; Katsigiannis and Ramzan 2018; Ab. Aziz et al. 2023) without solely relying on their self-evaluation. For example, after the self-assessment of the recording of MGEED (Wang et al. 2023), two researchers who had done professional training, were asked to verify the annotations independently at the frame level. Participants when asked to express themselves emotively, some subjects performed pretty well while others failed to express the requested emotions. Other subjects manage to perform satisfyingly, but only after several trials (Martin et al. 2006). Similarly, in (Wu and Liang 2011), each speaker of an utterance was asked to annotate the corresponding sentence manually. Another human subjection evaluation (annotator) was involved to annotate the data. In the case of the difference between perceived and intended emotions, the final annotation was determined according to the discussion between the annotator and the speaker. For IEMOCAP, the annotation of the categorical emotion was assigned based on the agreements derived from subjective emotional evaluations. Fluent English speakers were asked to evaluate the emotional content of the dataset. The self-assessment manikins (SAMs) (Bradley and Lang 1994) is a non-verbal pictorial assessment technique that directly measures the pleasure, arousal, and dominance associated with a person's affective reaction to a wide variety of stimuli. It was used for self-assessment rating in (Busso et al. 2008; Ringeval et al. 2013; Wang et al. 2023; Miranda et al. 2022; Velana et al. 2017; Ringeval et al. 2013). For example, in (Busso et al. 2008) SAMs (Bradley and Lang 1994) were used to evaluate the corpus in terms of the attributes valence [1-negative, 5-positive], activation [1-calm, 5-excited], and dominance [1-weak, 5-strong]. Then, two different evaluators were involved to evaluate the emotional content of the dataset using the SAM system. Daydic-based conversational datasets required three types of annotations: self-annotation, partner-annotation, and external observer annotation such as K-EmoCon (Park et al. 2020). In the annotation task described by Pérez Rosas et al. (2013a), ExpoTV.com's direct ratings were employed. For instance, in a 5-star movie review video, the speaker typically aimed to persuade the audience in favor of the movie, while in a 1-star movie review video, the speaker would argue against watching the movie.

Several tools are, also, used to manually annotate samples such as Amazon Mechanical Turk (Mason and Suri 2012) used in (Borth et al. 2013; You et al. 2016; Pereira et al. 2019; Poria et al. 2018; Zadeh et al. 2018; Nojavanasghari et al. 2016), ELAN (Brugman et al. 2004) used in (Pérez-Rosas et al. 2013; Nojavanasghari et al. 2016; Schimmel et al. 2019; Doyran et al. 2021), VADER (Hutto and Gilbert 2014) was applied to weakly annotate tweets in (You et al. 2016), AffectButton (Broekens and Brinkman 2013) was applied in (Boccignone et al. 2017), and FEELtrace system in (McKeown et al. 2010, 2011).

Annotation might be based on the whole video or on the utterance after segmentation. However, an utterance-level labeling scheme is particularly important to track the emotion and sentiment dynamics of the subject's mindset in a video (Poria et al. 2017). Annotation should be conducted based on different modalities. For example, the authors in (Wu and Liang 2011) found that a few data of annotating audio utterances for natural and sad are ambiguous but can be classified by textual transcription. In (Ellis et al. 2014), the authors demonstrated that up to 21.54% of sentiment annotations for transcripts deviate from their corresponding sentiment annotations when the actual video clip is presented. This finding is corroborated by (Poria et al. 2018).

In the case of extended unimodal datasets into multimodal datasets, it is recommended to re-annotate the extended dataset. For example, MELD (Poria et al. 2018) is the extended version of EmotionLines (Chen et al. 2018). The utterances in the original EmotionLines dataset were annotated by looking only at the transcripts. However, all utterances were re-annotated by asking the three annotators to also look at the available video clip of the utterances. This results in improving the annotation notably. The number of disagreements in MELD is 89, which is much lower than the 2,772 disagreements in EmotionLines (Chen et al. 2018). In addition, an overall Fleiss' kappa score of 0.43 for MELD was reported which is higher than Fleiss' kappa score of EmotionLines which is 0.34.

Several scores and statistics have been applied to measure the quality of the annotations including Kappa score (McHugh 2012), Finn's metric (Colorado and Eberle 2012), SScore (Bousmalis et al. 2011), Krippendorff's alpha statistic (Krippendorff 1970), and Spearman. Krippendorff's alpha statistic (Krippendorff 1970) establishes two points on the reliability scale: 1.0 denotes perfect reliability, 0.0 indicates the absence of reliability, and $\alpha < 0$ signifies systematic disagreements exceeding what can be anticipated by chance. Krippendorff's alpha was used to estimate the amount of inter-rater agreement in RAMAS (Perepelkina et al. 2018) dataset. Inter-annotator agreement measures the level of consistency among two or more annotators/evaluators in making identical annotation decisions for a given category (Artstein 2017).

In Liris-accede (Baveye et al. 2015), the Cronbach alpha coefficient was also used to measure the inter-evaluator agreement of the attribute-based evaluation with reporting 0.809, 0.607, and 0.608 for Valence, Activation, and Dominance, respectively.

5.3.6 Model training and deployment challenges:

In addition to the issues discussed above, there are several other challenges that can arise during model training and deployment due to the inherent complexity and heterogeneity of different modalities in emotional AI. Among the key challenges is modality representation and fusion to maximize complementary information derived from each modality. Choosing between early fusion (feature-level), late fusion (decision level), or hybrid fusion (integration of early and late) is crucial since improper fusion may lead to loss of critical emotional clues (Xue and Marculescu 2023). Compared to late fusion, early fusion can enable the model to effectively capture cross-modal interactions, it suffers from computational overhead and more prone to overfitting. In contrary, hybrid approaches can be more effective in capturing both independent and shared emotional representations. Other challenges include model convergence, stability, model overfitting and robust generalization, optimization bias, computational efficiency and co-learning complexities across modalities, model interpretability and explainability, hyperparameter tuning and loss balancing, defining suitable evaluation and benchmark metrics, knowledge distillation, model compression, deployment costs, and real-word usability (Middya et al. 2022; Baltrušaitis et al. 2018a; Kakuba et al. 2023; Gao et al. 2020). The significance of these issues depends on the specific application and context of emotional AI. For example, in high-stakes applications such as healthcare, model interpretability and explainability is most crucial. In contrast, for real-world environments, generalization will be most critical. Similarly, in edge computing and mobile applications, lightweight and model compression will be most critical.

5.3.7 Multi-objective datasets

A dataset with multiple objectives means that each data point is associated with more than one target or outcome variable. Creating multi-objective datasets is challenging due to several inherent complexities associated with capturing diverse and interconnected outcomes (Dembczyński et al. 2012; Zhang and Zhou 2013). Most of the reviewed datasets were created for single objective or target such as sentiment analysis such as YouTube (Morency et al. 2011), ICT-MMMO (Wöllmer et al. 2013), MOUD (Pérez-Rosas et al. 2013b), SMOD (Pérez Rosas et al. 2013a), (Pérez Rosas et al. 2013a), News Rover (Ellis et al. 2014) or emotion detection such as Belfast (Douglas-Cowie et al. 2000), EmoTV1 (Abrilian et al. 2005), eINTERFACE (Martin et al. 2006), HUMAINE (Douglas-Cowie et al. 2007), (Caridakis et al. 2007), IEMOCAP (Busso et al. 2008), VAM (Grimm et al. 2008), SEMAINE (McKeown et al. 2010). Other datasets were constructed for more than one objective or target such as ECSMP (Gao et al. 2021) for emotion, cognition, and sleep quality, SWELL-KW (Koldijk et al. 2014) for emotion and stress, SADAM (Al-Azani and El-Alfy 2020) for sentiment and demographics, MOSEI (Zadeh et al. 2018) and MELD (Poria et al. 2018) for emotion and sentiment.

Some studies originally were annotated for emotion states then extended to include sentiments through converting fine-grained emotion into more coarse-grained sentiment classes (Poria et al. 2018). In (Poria et al. 2018), anger, disgust, fear, and sadness were considered as negative while joy was considered as positive. However, the surprise utterances were re-annotated as either positive or negative sentiment classes. However, the diversity and complexity of the domain make it challenging to provide standard recordings that adequately address the majority of potential applications and research questions (Lukowicz et al. 2010)

6 Emerging Directions

As emotional AI continues to advance, emerging innovations in multimodal datasets are enhancing emotion recognition, fostering greater accuracy, and real-world applicability. Some of the significant aspects are discussed below.

A) Role of Large-Scale Vision and Language Models:

With the advances in AI, several approaches have been proposed in the literature to leverage the emerging Large-Scale Vision and Language models (LVLMs), such as DeepSeek-VL, GPT-4V, BLIP-2, and Flamingo. LVLMs enable processing of multiple modalities and shifting from task-driven to AI-assisted emotional-aware systems, not just for emotion recognition, but also for contextual understanding, adaptive reasoning, dynamic response generation, and real-time personalized interaction. Unlike traditional emotion detection systems that simply classify affective states (e.g., "happy", "angry") or sentiment polarities ("positive", "negative", "neutral"), next-generation emotional AI requires datasets that capture the full spectrum of affective interaction, including emotional expression and intent (Guo et al. 2024; Otterdijk et al. 2024), social-emotional context (Henriksen et al. 2025), among others. Subsequently, there is a need for large-scale open-source multimodal emotion datasets for training and benchmarking innovative ideas and systems across multiple modalities. As an example, in (Lian et al. 2024) described a large-scale dataset and a model for Explainable

Multimodal Emotion Recognition. The dataset is composed of 115,595 samples and the AffectGPT model is a two stage framework which first learns a coarse mapping and refines it to better align with manually-checked results. Another example is SAMSEMO (Bujnowski et al. 2024), which contains over 23 thousands open-source video scenes in five languages (English, German, Spanish, Polish, and Korean). Additionally, LVLMs can help to relieve the burden of data annotation and augmentation, mitigate data scarcity, provide better performance for broader generalization and transferability across applications and cross-cultural environments.

B) Synthetic Data in Addressing Data Scarcity:

As mentioned previously, deep learning techniques require a lot of data in general and specifically for emotional data due to the need for diverse and nuanced examples to accurately capture the wide range of human emotions, expressions, and contexts. Without sufficient data, models may struggle to generalize, leading to poor performance and biased results. Synthetic data can have a significant impact in mitigating such issues by generating diverse, balanced, and contextually rich emotional datasets. This enables deep learning models to train on a broader range of emotional expressions and scenarios, helping to overcome data scarcity and enhance model accuracy, robustness, and generalization. In general, besides traditional data augmentation techniques, there are several approaches including generative adversarial networks and variational autoencoders to generate synthetic data, whether unimodal or multimodal. Additionally, more advanced methods for data generation includes diffusion models and Large-Scale Vision-Language Models have been introduced, e.g. (Ma et al. 2024; Xu et al. 2024). While synthetic data is a valuable tool for overcoming data scarcity, its reliability depends on how well it represents real-world conditions and how it is validated. It is best used as a supplement to real data, rather than a complete replacement (Wiehn 2024).

C) Emerging Trends in Transformer-Based Models for Multimodal Fusion:

In the literature, multimodal data can be fused at various levels including feature level, score level and decision level. Though these technique have simple to moderate levels of complexities, they may be limited in capturing complex interactions between modalities and may lack robustness. Transformer-based deep learning models have recently led to a revolution in AI for effective integration of text, speech and visual modalities for various tasks of language processing and computer vision (Xu et al. 2023; Khan et al. 2025). A comprehensive review and taxonomy of transformer-based models for multimodal emotion recognition is presented by Hazmoune and Bougamouza (2024).

D) Emerging Applications of Emotional AI:

There are many real-world applications across various domains, including social robotics and assistive technology. Recent advancements in deep learning and generative AI have significantly enhanced affective computing, enabling robots not only to recognize and interpret human emotions, but also to generate and express emotional cues with greater accuracy and cognitive granularity. Another major trend is the application in emotion-aware online and adaptive leaning environments, elderly care, health monitoring, and personalized treatment (Vistorte et al. 2024; Ahuja 2024).

7 Conclusions

Research on multimodal datasets for emotional AI has expanded significantly in recent years. This review article has thoroughly addressed several critical research aspects and directions related to multimodal emotional AI datasets and their applications. The exploration of major trends in the field has shed light on the diverse tasks that leverage the development of more effective multimodal machine learning approaches for understanding and interpreting emotions. The key characteristics of available multimodal datasets have been thoroughly analyzed. The paper also highlighted the intricate process involved in the development and creation of related multimodal datasets and outlined recommendations for developers of these datasets. By answering questions related to dataset construction processes, challenges, and solutions, this review contributes to a better understanding of the complexities of creating more reliable and diverse multimodal datasets. The challenges associated with developing appropriate datasets were being actively addressed by existing initiatives, emphasizing the importance of collaboration and standardization in this domain. Moreover, the paper has delved into various concerns and ethical considerations surrounding multimodal datasets. Overall, dataset developers should prioritize creating standardized protocols and tools for data collection and processing to ensure completeness, consistent alignment, synchronization, and reliability and validity of annotations across different modalities. It is crucial to adhere to ethical standards and adopt clear guidelines and policies to safeguard participant's privacy and rights. Furthermore, they should provide comprehensive documentation to facilitate model development, promote transparency, and enhance reproducibility of research results. Understanding those issues is crucial for next generation advancing the research in the domain of emotional AI and development of next generation systems. This review serves as an intensive resource for the research community interested in assessing the current research trends and available resources for making further advances in the field of emotional AI and diverse related applications. As future work, a deeper comparative analysis of various benchmarking datasets used in emotion recognition tasks can be conducted. Moreover, as another direction is to develop standardized frameworks, protocols, and evaluation metrics for emotion recognition tasks and to assess cross-dataset performance of various models.

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